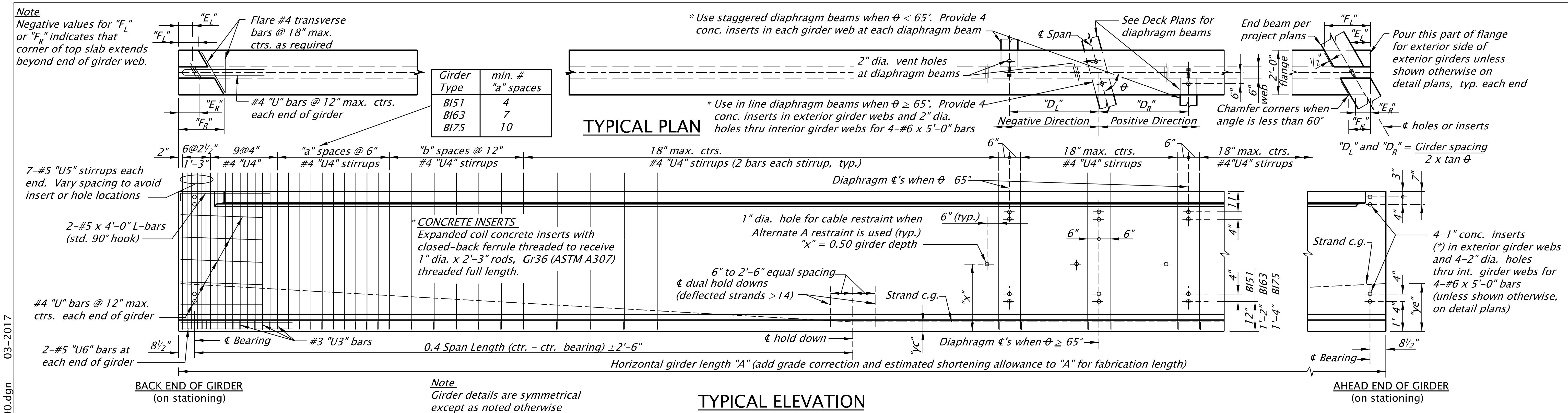
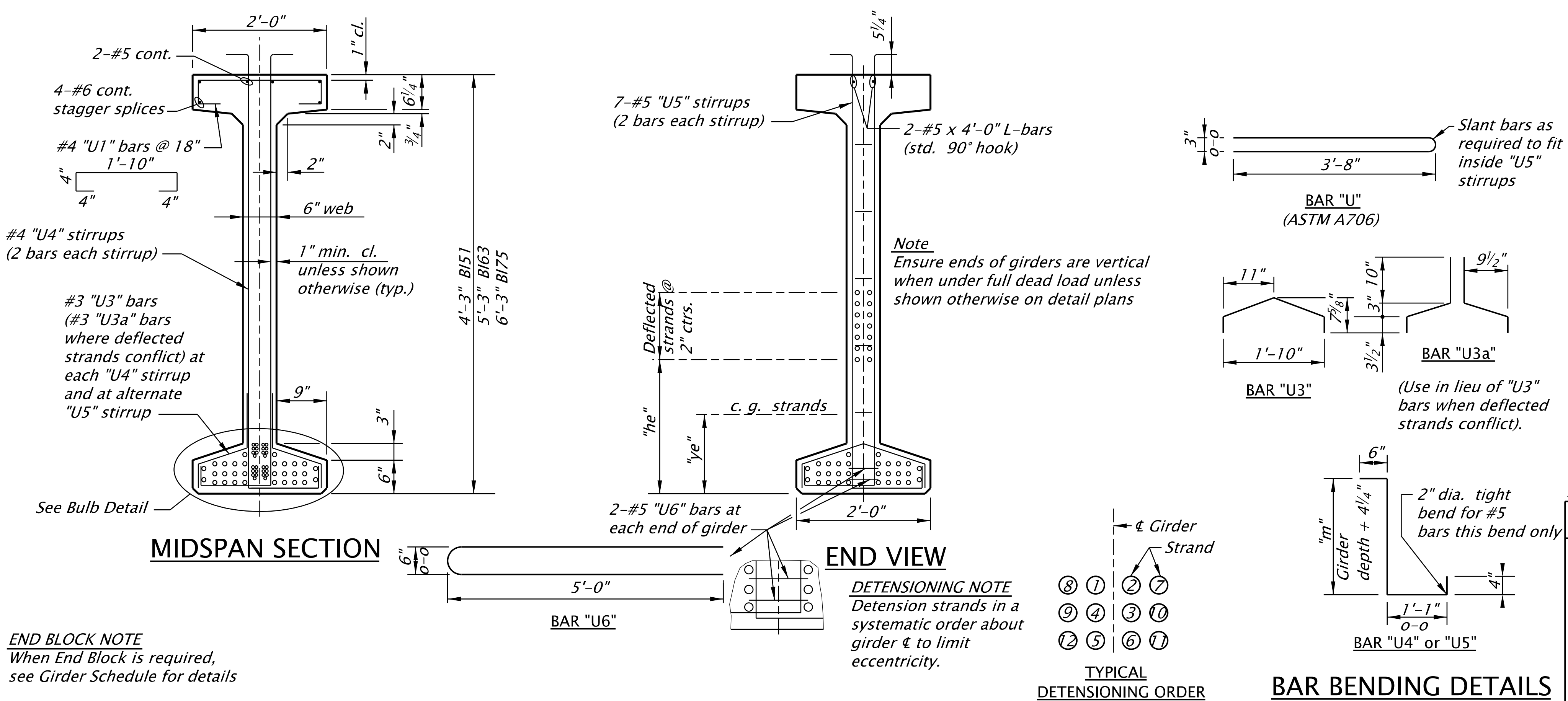
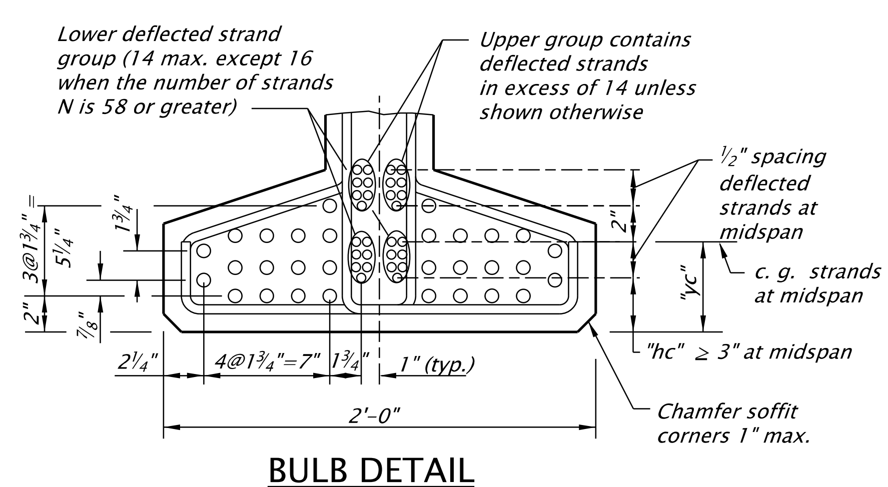


Girder Type	min. # "a" spaces
BI51	4
BI63	7
BI75	10





END BLOCK NOTE
When End Block is required, see Girder Schedule for details

BULB-I GIRDER NOTES
Girders designed for live loading and superimposed dead load as shown in the General Notes.
Provide reinforcing steel as shown in the General Notes except as noted.
Provide ½" dia. 7-wire low relaxation prestressing strands conforming to AASHTO M203 (ASTM A416), Grade 270, Supplement 1. Provide an initial tensioning (after harping deflected strands) of 31.0 kips per strand unless shown otherwise on the Girder Schedule.

Provide the class of concrete shown on the Girder Schedule.
Keep girders in an upright position with the support points within 2'-0" of the girder ends.
Deflected strands may be bundled between hold down points.
Provide temporary diaphragm beams as required, see BR350.

Accompanied by dwgs. BR350 and DET3305

CALC. BOOK NO. -

SDR DATE: 20-April-2018

NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.

OREGON STANDARD DRAWINGS

BULB-I (PRECAST PRESTRESSED CONCRETE GIRDERS)

2022

DATE	REVISION DESCRIPTION
-	-
-	-
-	-
-	-
-	-

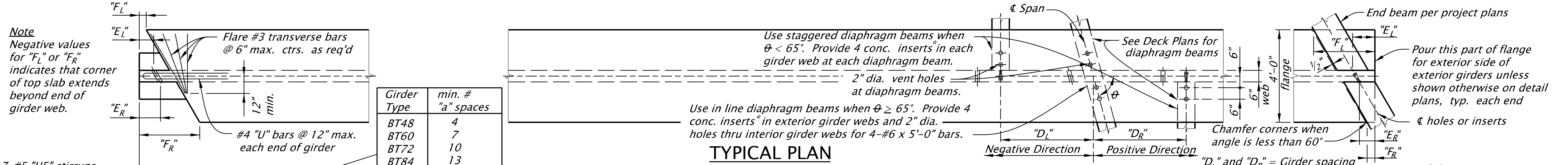
03-2017

br300.dgn

BR300

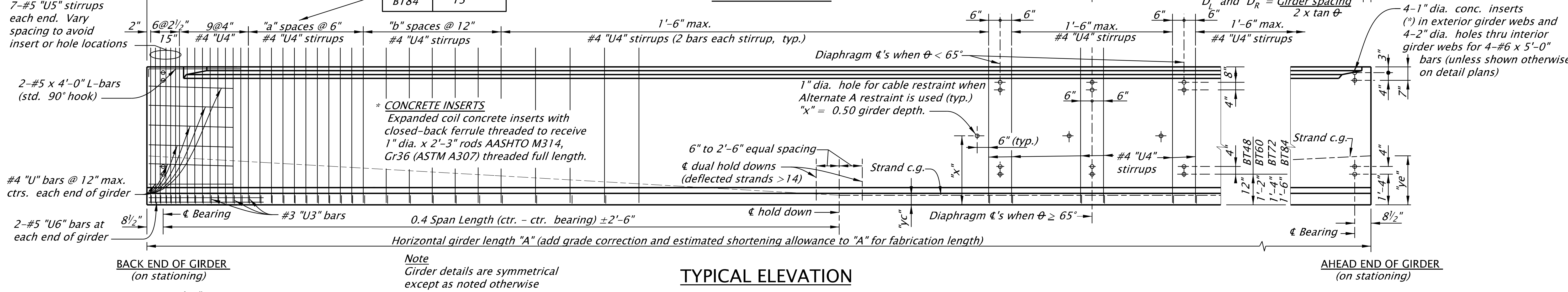
Effective Date: December 1, 2022 - May 31, 2023

BR300



Girder Type	min. # "a" spaces
BT48	4
BT60	7
BT72	10
BT84	13

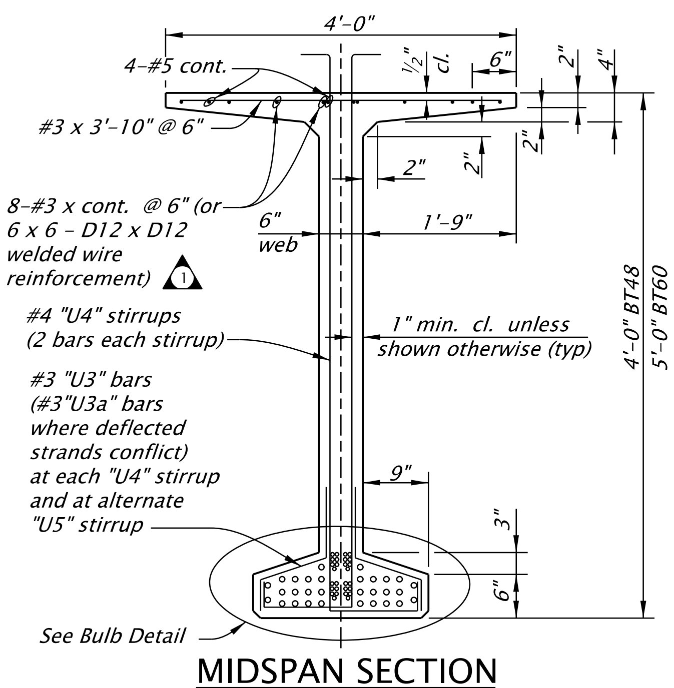
TYPICAL PLAN



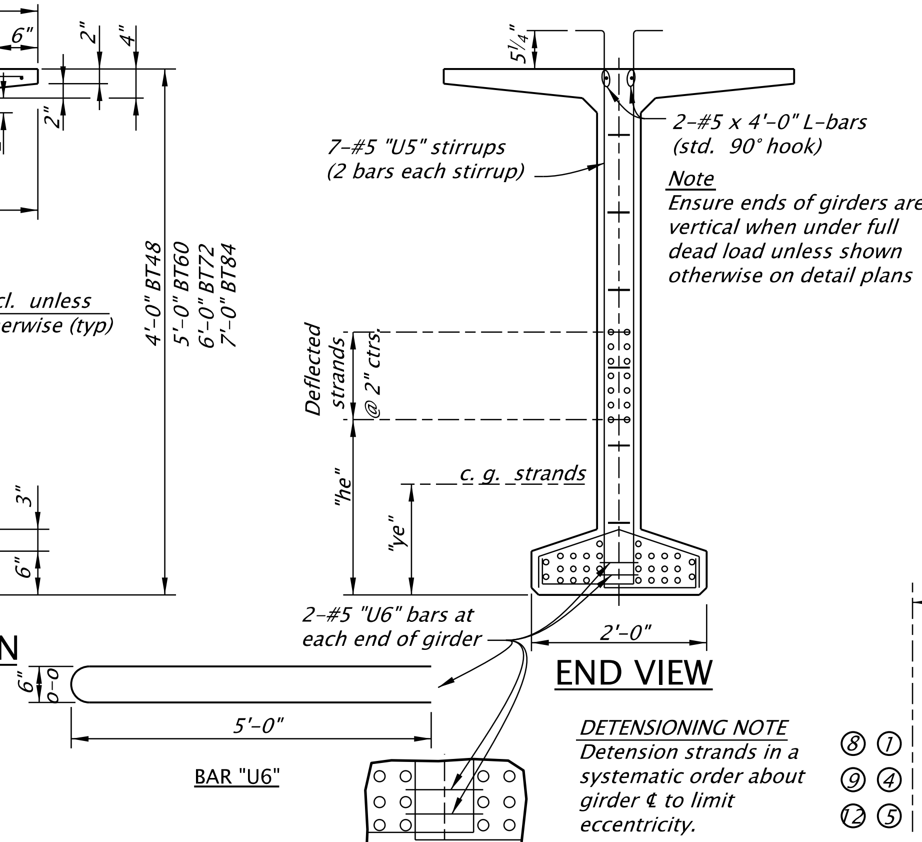
* CONCRETE INSERTS
Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods AASHTO M314, Gr36 (ASTM A307) threaded full length.

Note
Girder details are symmetrical except as noted otherwise

TYPICAL ELEVATION



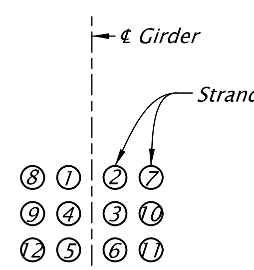
MIDSPAN SECTION



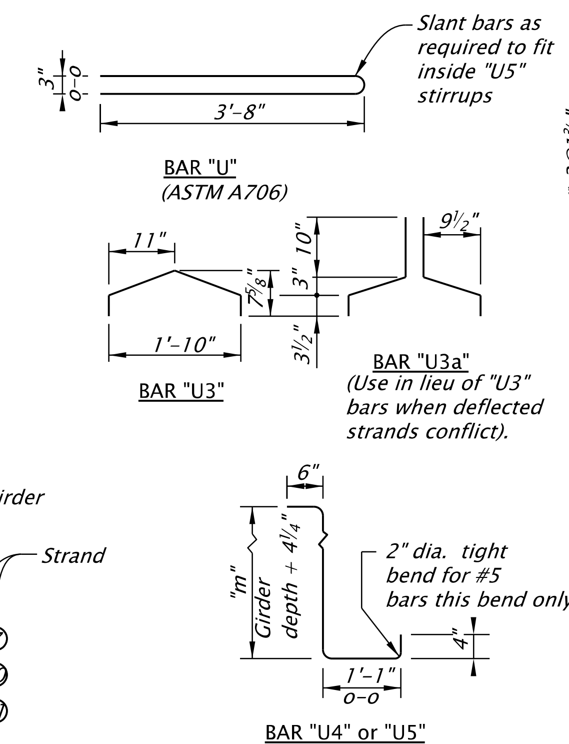
END VIEW

END BLOCK NOTE
When End Block is required, see Girder Schedule for details

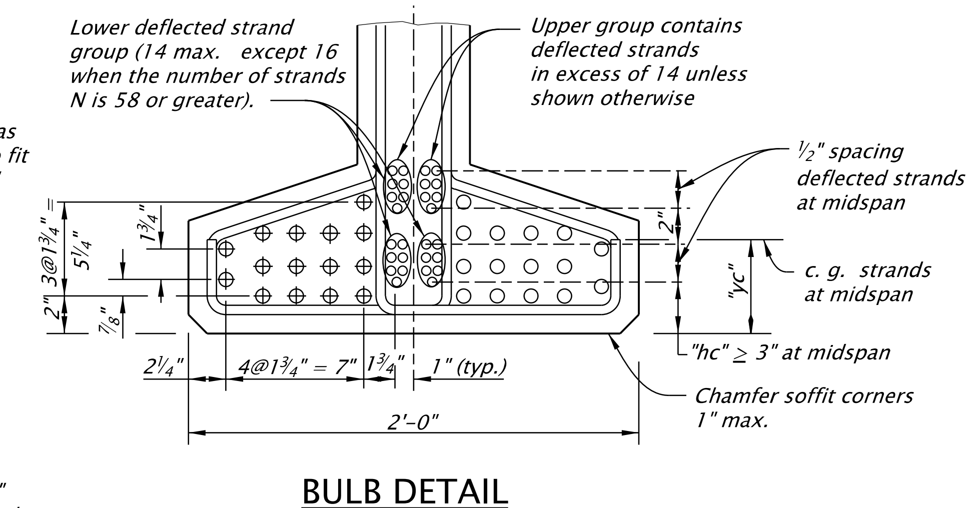
DETENSIONING NOTE
Detension strands in a systematic order about girder ¢ to limit eccentricity.



TYPICAL DETENSIONING ORDER



BAR BENDING DETAILS



BULB DETAIL

BULB-T GIRDER NOTES
Girders designed for live loading and superimposed dead load as shown in the General Notes.
Provide reinforcing steel as shown in the General Notes except as noted.
Provide welded wire reinforcement conforming to ASTM A1064. (1)
Provide ½" dia. 7-wire low relaxation prestressing strands conforming to AASHTO M203 (ASTM A416), Grade 270, Supplement 1. Provide an initial tensioning (after harping deflected strands) of 31.0 kips per strand unless shown otherwise on Girder Schedule.

Provide the class of concrete shown on Girder Schedule.
Keep girders in an upright position w/the support points within 2'-0" of the girder ends.
Deflected strands may be bundled between hold-down points.
Provide temporary diaphragm beams as required, see BR350.

Accompanied by dwgs. BR350 and DET3315

CALC. BOOK NO. -

SDR DATE: 05-January-2022

NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.

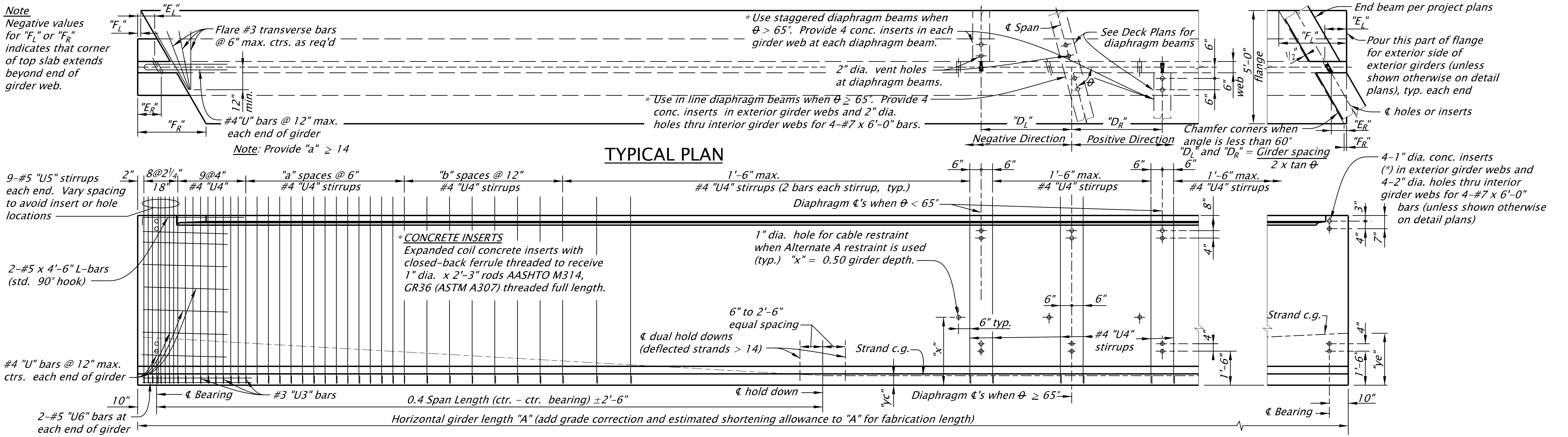
OREGON STANDARD DRAWINGS

BULB-T (PRECAST PRESTRESSED CONCRETE GIRDERS)

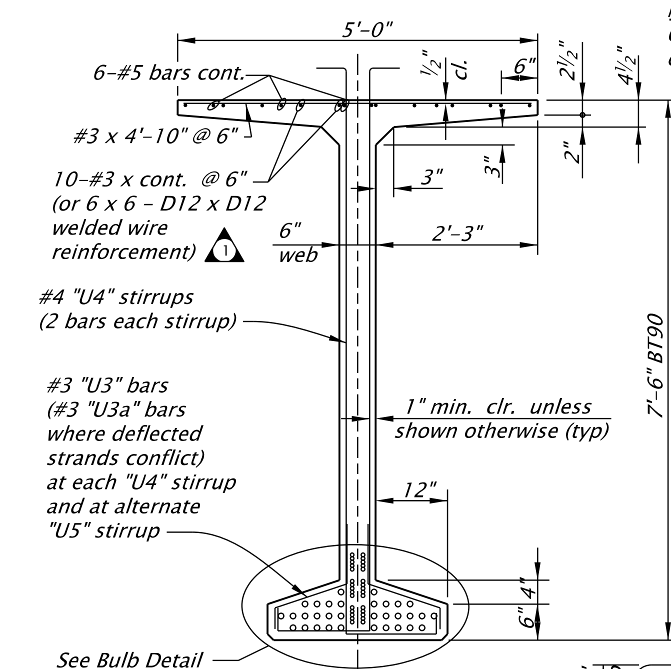
2022

	DATE	REVISION DESCRIPTION
1	01-2022	Revised welded wire reinforcement specification.
	-	-
	-	-
	-	-
	-	-

12/28/21
br310.dgn

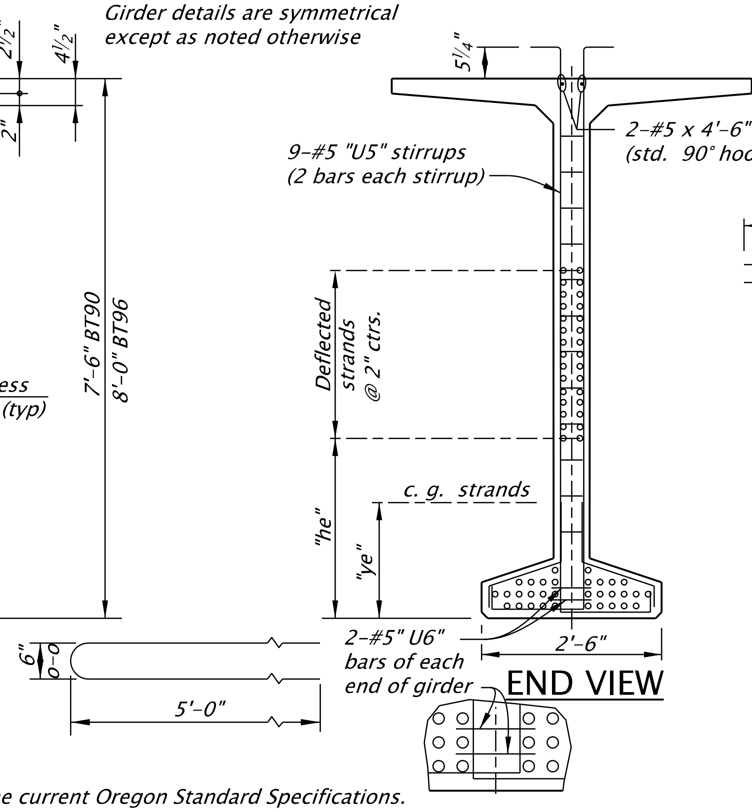


TYPICAL ELEVATION



MIDSPAN SECTION

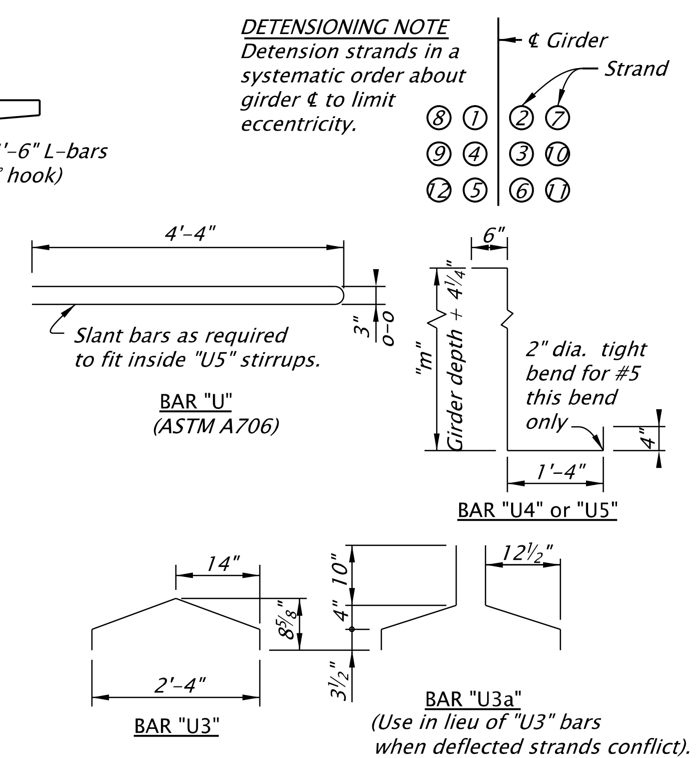
Note
Girder details are symmetrical except as noted otherwise



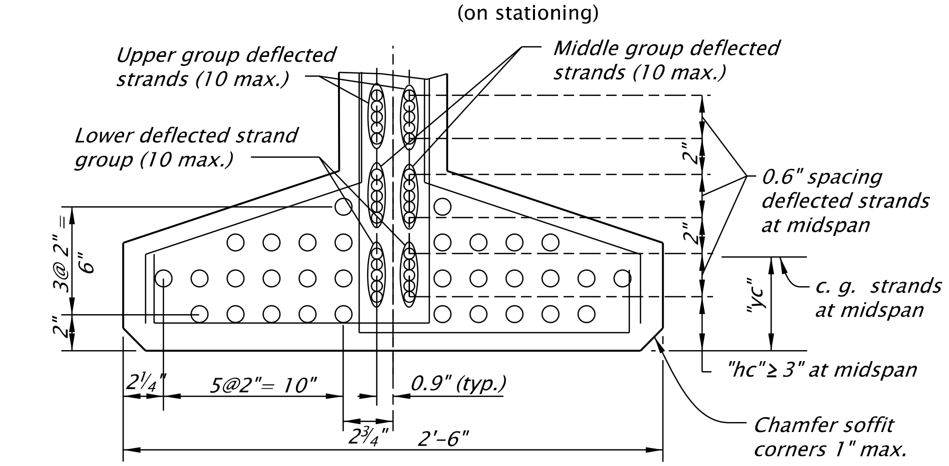
END VIEW

TYPICAL DETENSIONING ORDER

DETENSIONING NOTE
Detension strands in a systematic order about girder ℄ to limit eccentricity.



BAR BENDING DETAILS



BULB DETAIL

END BLOCK NOTE
When End Block is required, see Girder Schedule for details

NOTES
Provide all material and preform all work according to the current Oregon Standard Specifications.
Girders designed for live loading and superimposed dead load as shown in the General Notes.
Provide reinforcing steel as shown in the General Notes except as noted. Provide welded wire reinforcement conforming to ASTM A1064. ⚠1
Provide 0.6" dia. 7-wire low relaxation strands conforming to AASHTO M203 (ASTM A416), Grade 270, Supplement 1. Provide an initial tension (after harping deflected strands) of 43.9 kips per strand unless shown otherwise on Girder Schedule.
Provide the class of concrete shown on Girder Schedule.
Keep girders in an upright position w/the support points within 2'-0" of the girder ends.
Deflected strands may be bundled between hold-down points.
Provide temporary diaphragm beams as required, see BR350.
Ensure girder ends are plumb in their final position under full dead load.

Accompanied by dwgs. BR350 and DET3322

CALC. BOOK NO. -

SDR DATE: 05-January-2022

NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.

OREGON STANDARD DRAWINGS

BT90 AND BT96 (PRECAST PRESTRESSED CONCRETE GIRDERS)

2022

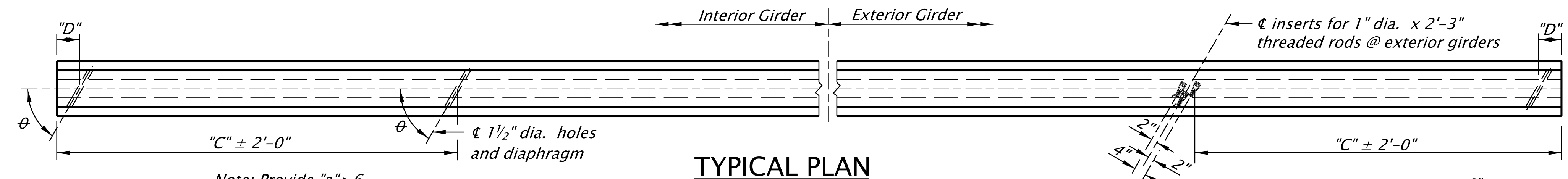
	DATE	REVISION DESCRIPTION
1	01-2022	Revised welded wire reinforcement specification.
	–	–
	–	–
	–	–
	–	–

br321.dgn 12/28/21

BR321

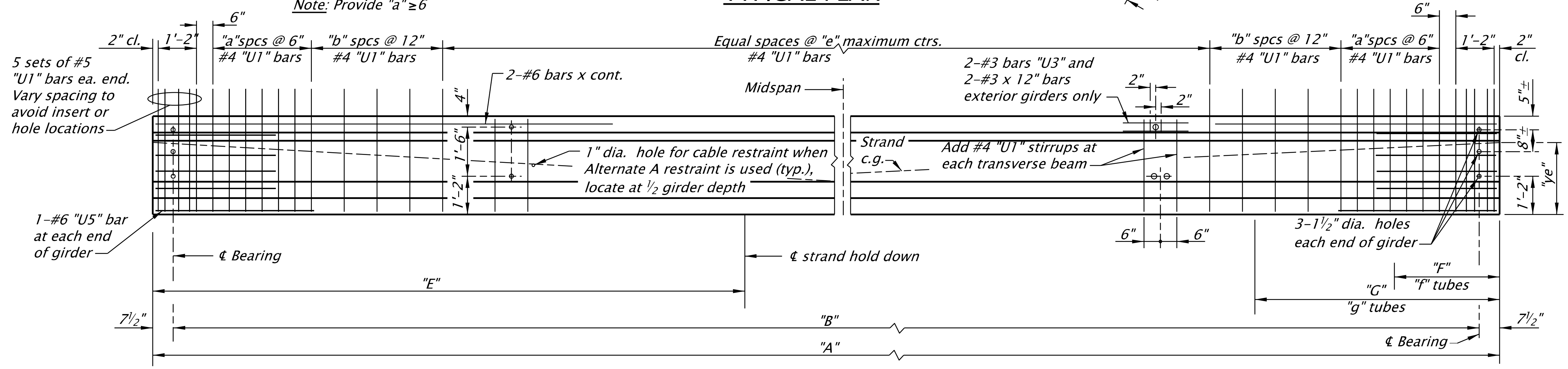
Effective Date: December 1, 2022 - May 31, 2023

BR321



TYPICAL PLAN

Note: Provide "a" ≥6



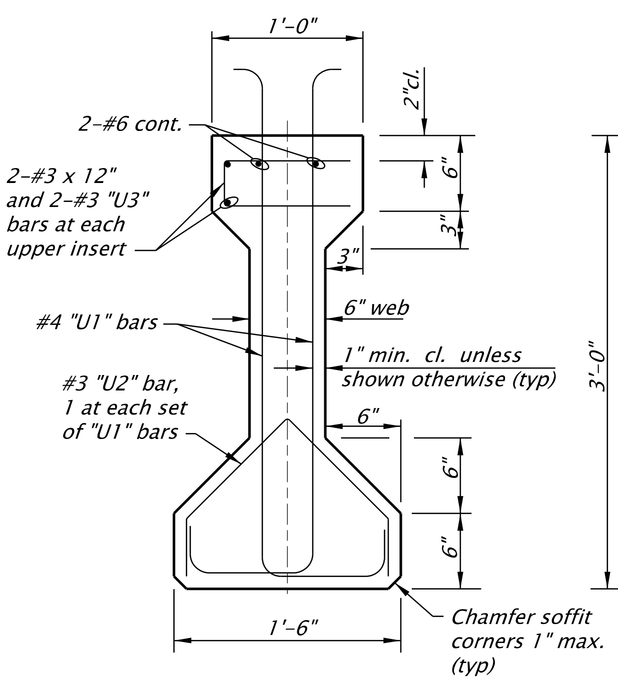
BACK END OF GIRDER
(on stationing)

TYPICAL ELEVATION

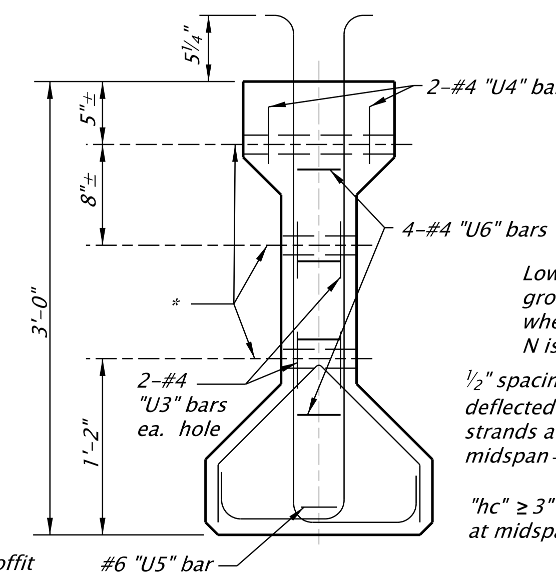
AHEAD END OF GIRDER
(on stationing)

Note
Girder details are symmetrical about midspan unless shown otherwise.

Note
These sections for dimensions only. See Figures on the Girder Schedule Sheet for the actual strand arrangements to be used.

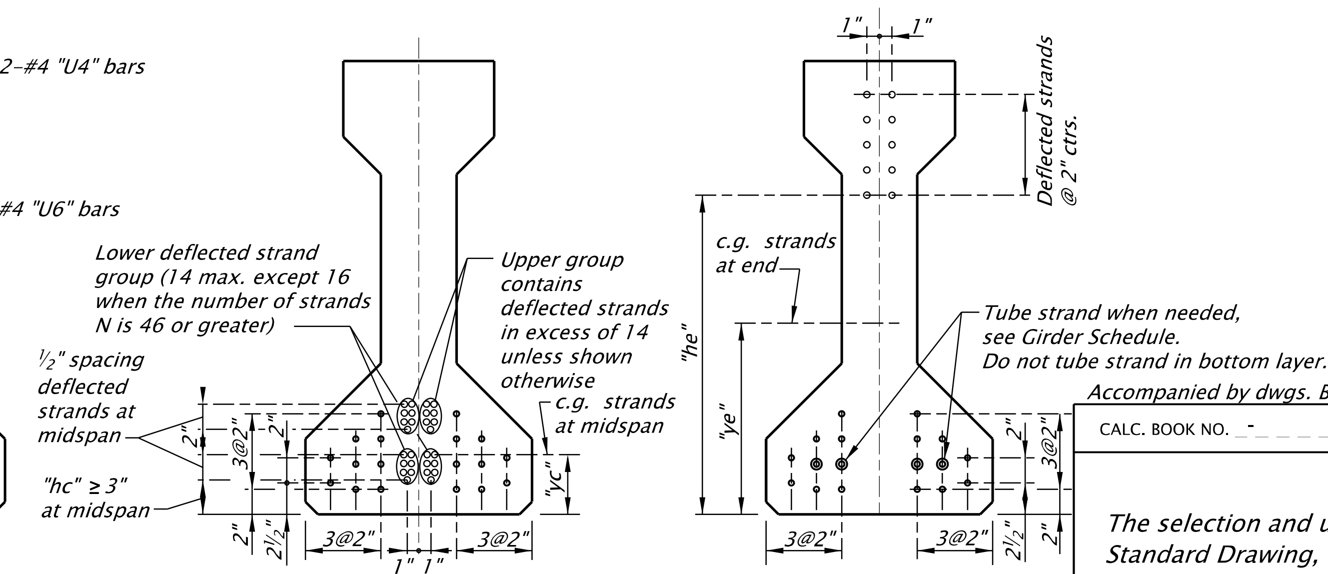


MIDSPAN SECTION



END VIEW

* ℄ 1½" dia. holes in internal girders.
Inserts in exterior girders.



MIDSPAN SECTION

END VIEW

STRAND ARRANGEMENTS

PRETENSIONING STEEL

Provide ½" dia. 7-wire low relaxation strands conforming to AASHTO M203 (ASTMA416), Grade 270, Supplement 1.
Provide an initial tension (after harping deflected strands) of 31.0 kips per strand unless shown otherwise. Deflected strands may be bundled between hold down points.
Provide the same tension as the straight strands after deflecting.
Detension strands alternately about the girder ℄ to limit prestress eccentricity.

MILD STEEL REINFORCEMENT

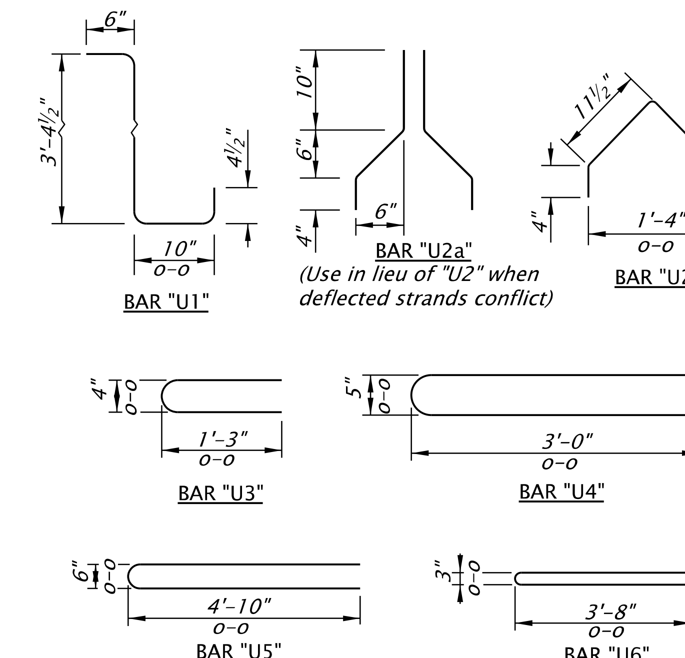
Provide reinforcing steel as specified in the General Notes for the project unless shown otherwise.

HANDLING PRESTRESSED CONCRETE GIRDERS

Keep girders in an upright position with the support points within 2'-0" of the girder ends.
Provide temporary diaphragm beams as required, see BR350.

DIAPHRAGM BEAMS

See structure plans for diaphragm beam locations.
Pour transverse beams a minimum of three days before deck pour.
Provide structural concrete inserts:
Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods AASHTO M314, Gr. 36 (ASTM A307) threaded full length.
Brace prestressed girders to prevent lateral deflection during diaphragm pour.



Accompanied by dwgs. BR350 and DET3345

CALC. BOOK NO. -	SDR DATE: 20-April-2018
	NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.

OREGON STANDARD DRAWINGS

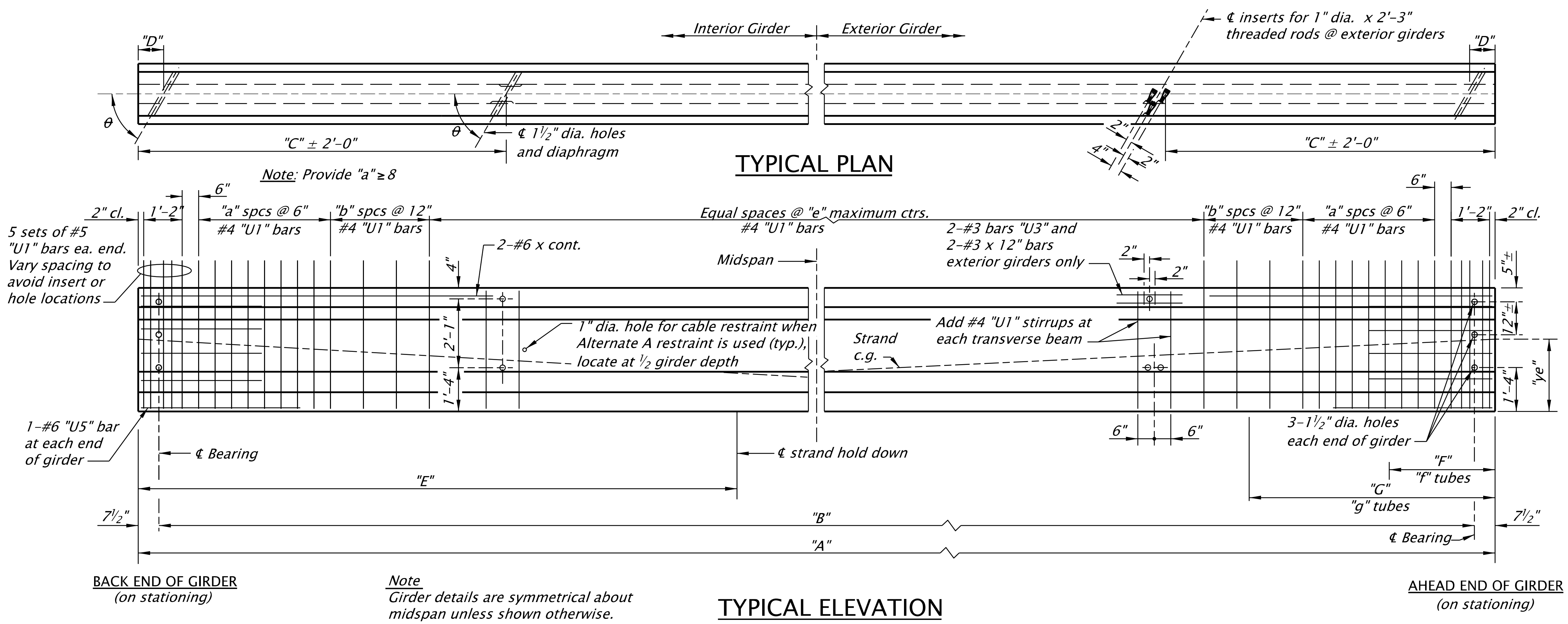
TYPE II - PRESTRESSED CONCRETE GIRDERS

2022

DATE	REVISION DESCRIPTION
-	-
-	-
-	-
-	-
-	-

br325.dgn 03-2017

BR325



PRETENSIONING STEEL

Provide ½" dia. 7-wire low relaxation strands conforming to AASHTO M203 (ASTMA416), Grade 270, Supplement 1.

Provide an initial tension (after harping deflected strands) of 31.0 kips per strand unless shown otherwise. Deflected strands may be bundled between hold down points.

Provide the same tension as the straight strands after deflecting.

Detension strands alternately about the girder ¢ to limit prestress eccentricity.

MILD STEEL REINFORCEMENT

Provide reinforcing steel as specified in the General Notes for the project unless shown otherwise.

HANDLING PRESTRESSED CONCRETE GIRDERS

Keep girders in an upright position with the support points within 2'-0" of the girder ends.

Provide temporary diaphragm beams as required, see BR350.

DIAPHRAGM BEAMS

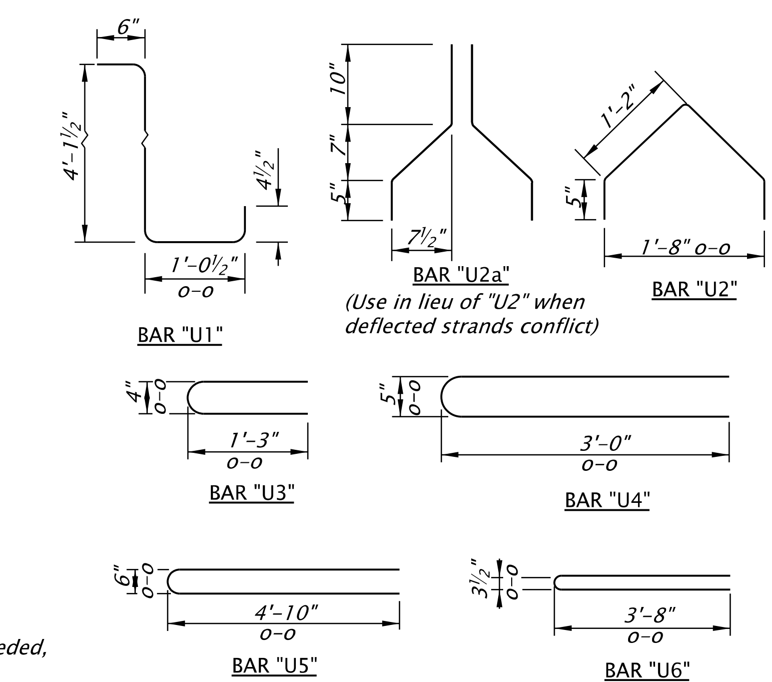
See structure plans for diaphragm beam locations.

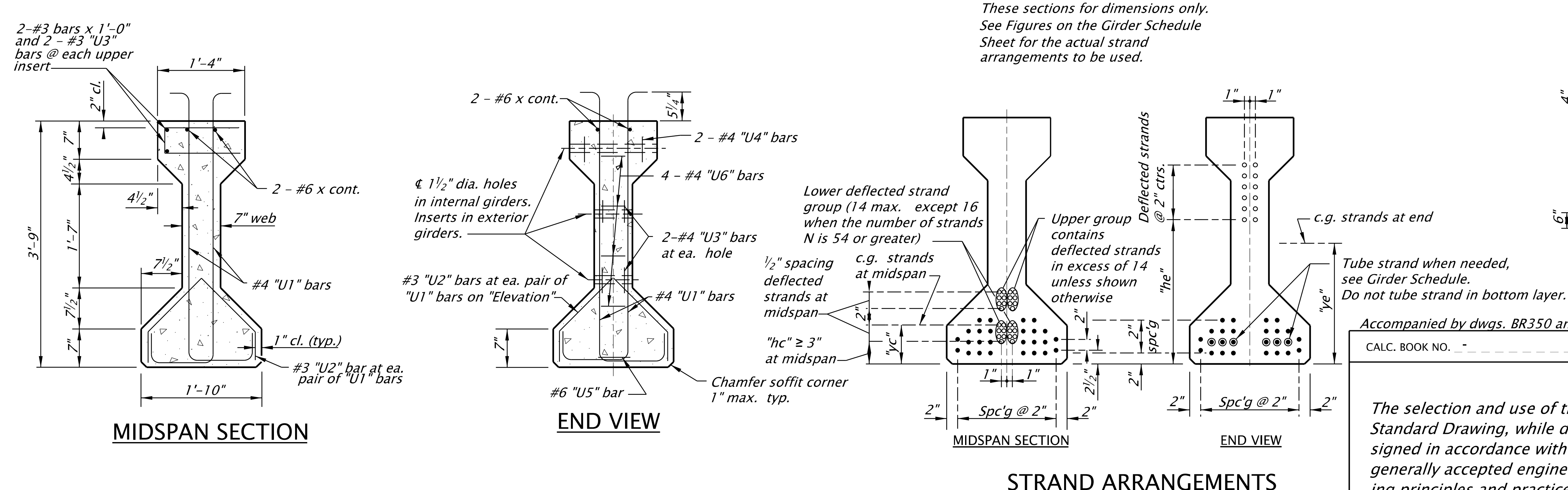
Pour transverse beams a minimum of three days before deck pour.

Provide structural concrete inserts:

Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods AASHTO M314, Gr. 36 (ASTM A307) threaded full length.

Brace prestressed girders to prevent lateral deflection during diaphragm pour.





Accompanied by dwgs. BR350 and DET3345

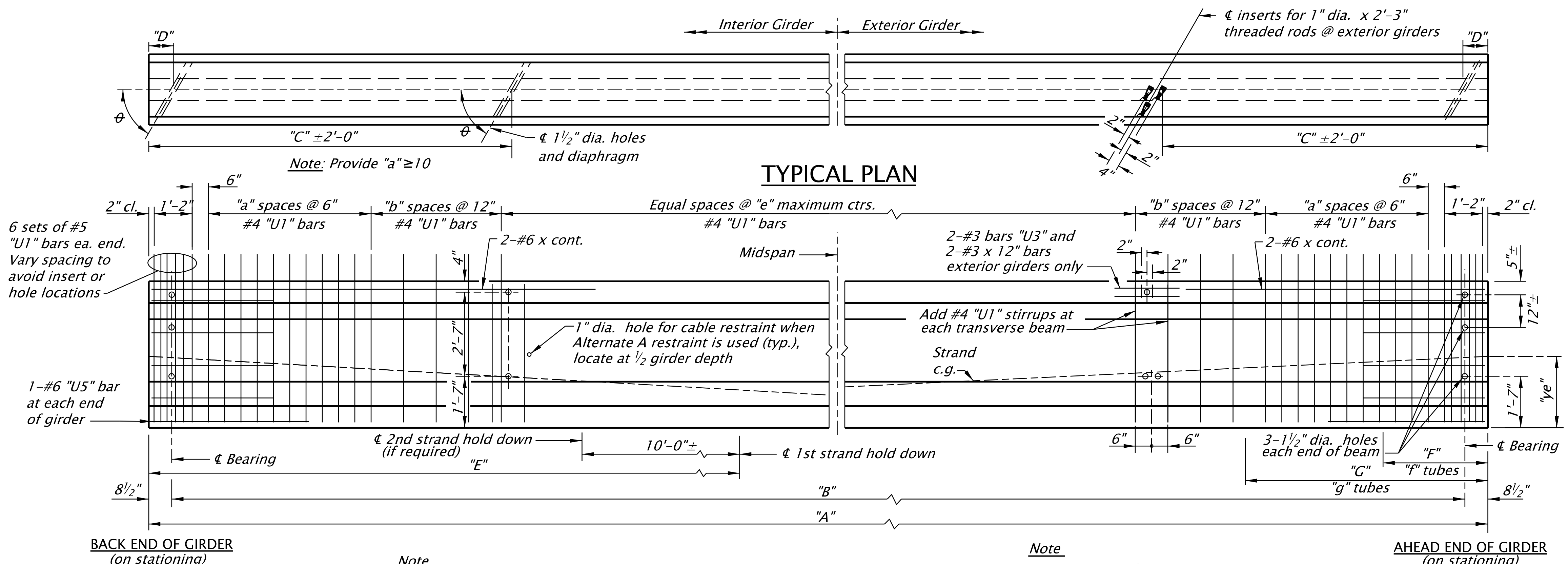
CALC. BOOK NO. -	SDR DATE: 20-April-2018
	NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications
The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.	**OREGON STANDARD DRAWINGS**

TYPE III - PRESTRESSED CONCRETE GIRDERS

2022

DATE	REVISION DESCRIPTION
-	-
-	-
-	-
-	-
-	-

BR330



TYPICAL PLAN

TYPICAL ELEVATION

Note
Girder details are symmetrical about midspan unless shown otherwise

Note
These sections for dimensions only. See Figures on the Girder Schedule Sheet for the actual strand arrangements to be used.

PRETENSIONING STEEL

Provide ½" dia. 7-wire low relaxation strands conforming to AASHTO M203 (ASTMA416), Grade 270, Supplement 1.

Provide an initial tension (after harping deflected strands) of 31.0 kips per strand unless shown otherwise. Deflected strands may be bundled between hold down points.

Provide the same tension as the straight strands after deflecting.

Detension strands alternately about the girder ¢ to limit prestress eccentricity.

MILD STEEL REINFORCEMENT

Provide reinforcing steel as specified in the General Notes for the project unless shown otherwise.

HANDLING PRESTRESSED CONCRETE GIRDERS

Keep girders in an upright position with the support points within 2'-0" of the girder ends.

Provide temporary diaphragm beams as required, see BR350.

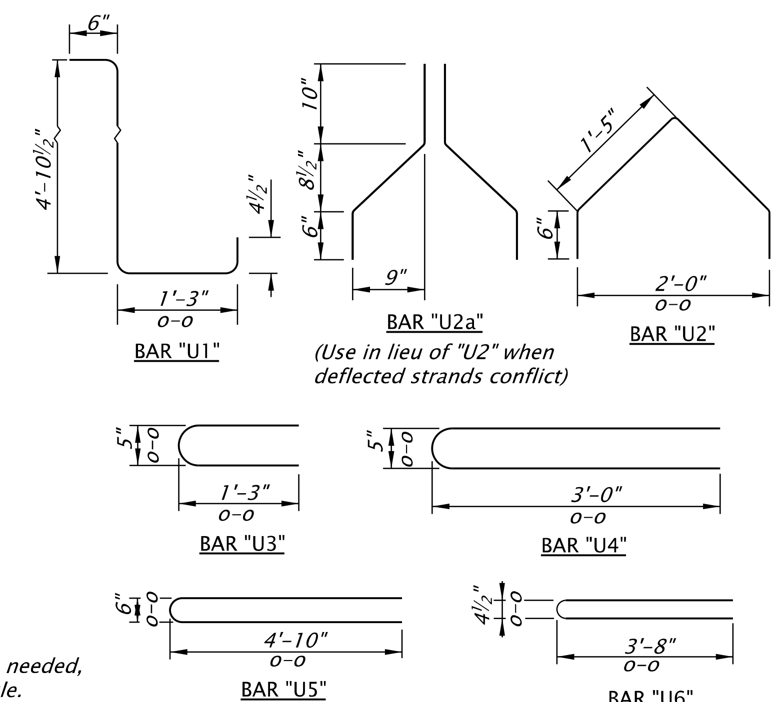
DIAPHRAGM BEAMS

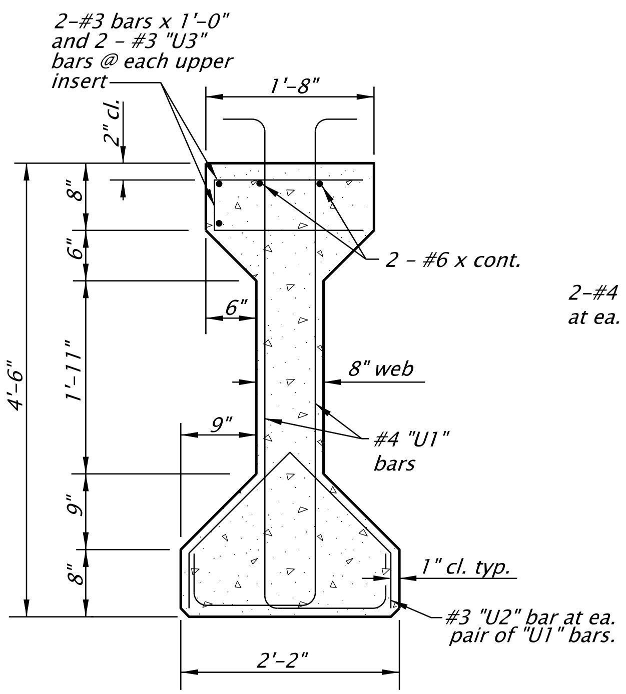
See structure plans for diaphragm beam locations.

Pour transverse beams a minimum of three days before deck pour.

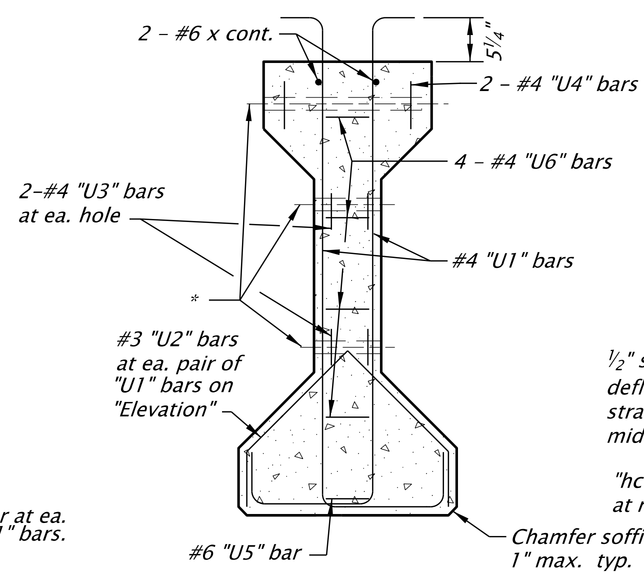
Provide structural concrete inserts:
Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods AASHTO M314, Gr.36 (ASTM A307) threaded full length.

Brace prestressed girders to prevent lateral deflection during diaphragm pour.



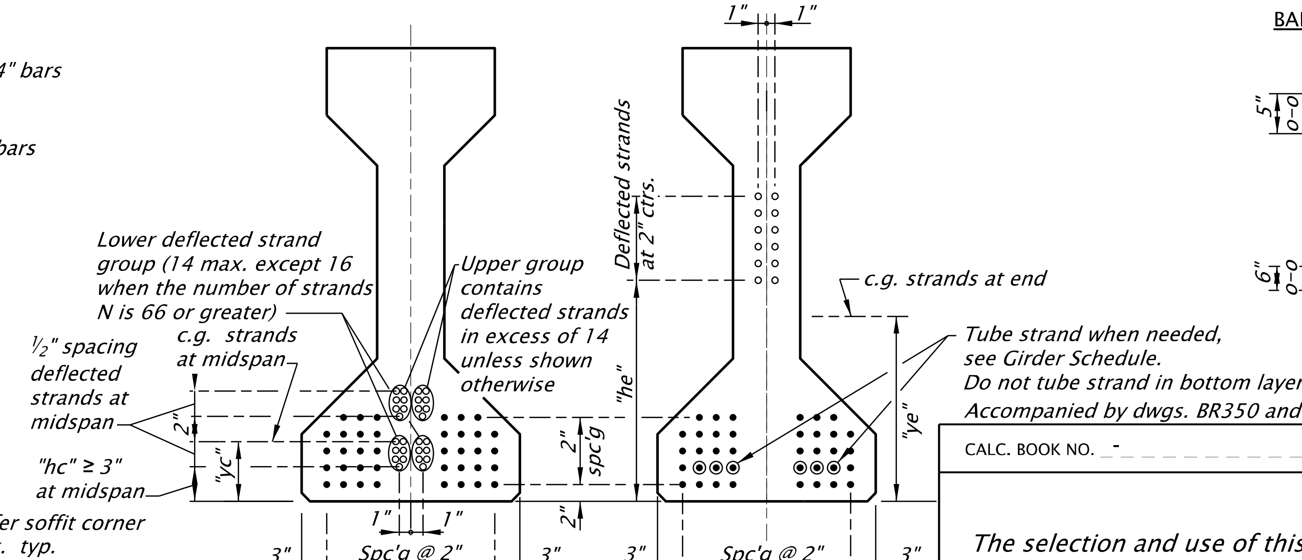


MIDSPAN SECTION



END VIEW

* ¢ 1½" dia. holes in internal girders. Inserts in exterior girders.



STRAND ARRANGEMENTS

CALC. BOOK NO. -

SDR DATE: 20-April-2018

NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.

OREGON STANDARD DRAWINGS

TYPE IV - PRESTRESSED CONCRETE GIRDERS

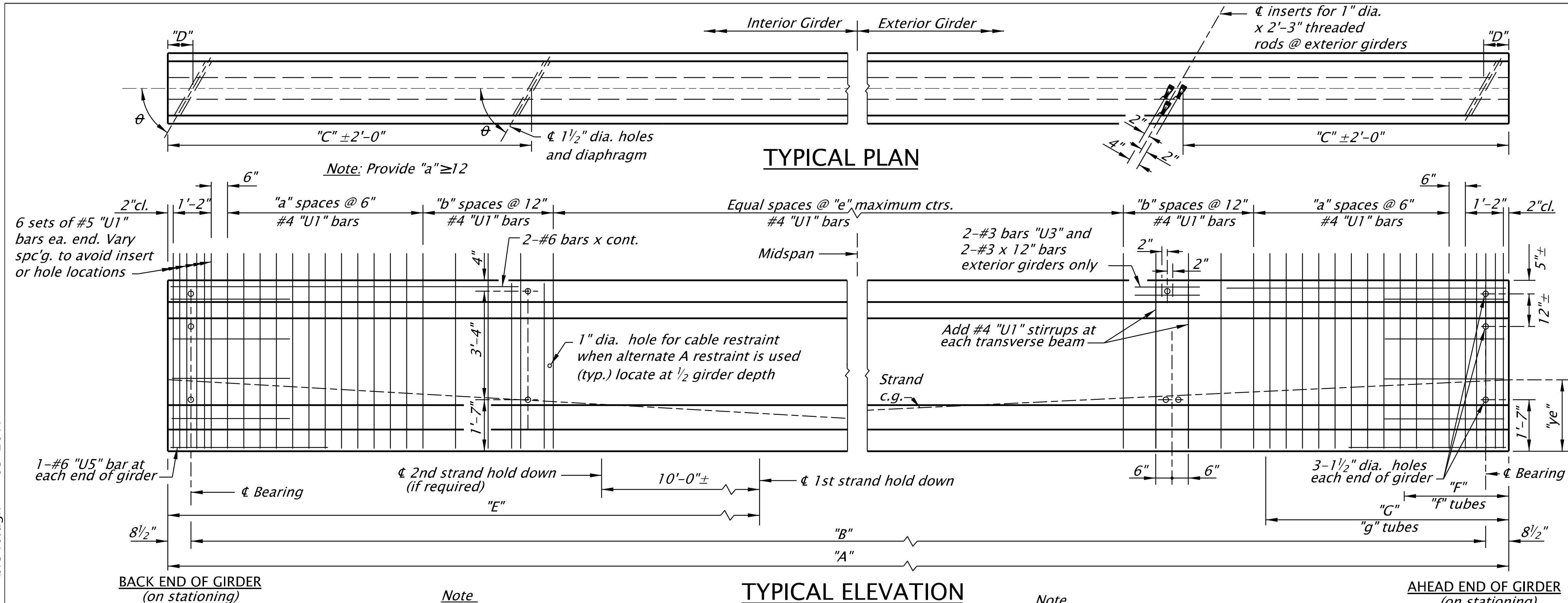
2022

DATE	REVISION DESCRIPTION
-	-
-	-
-	-
-	-
-	-

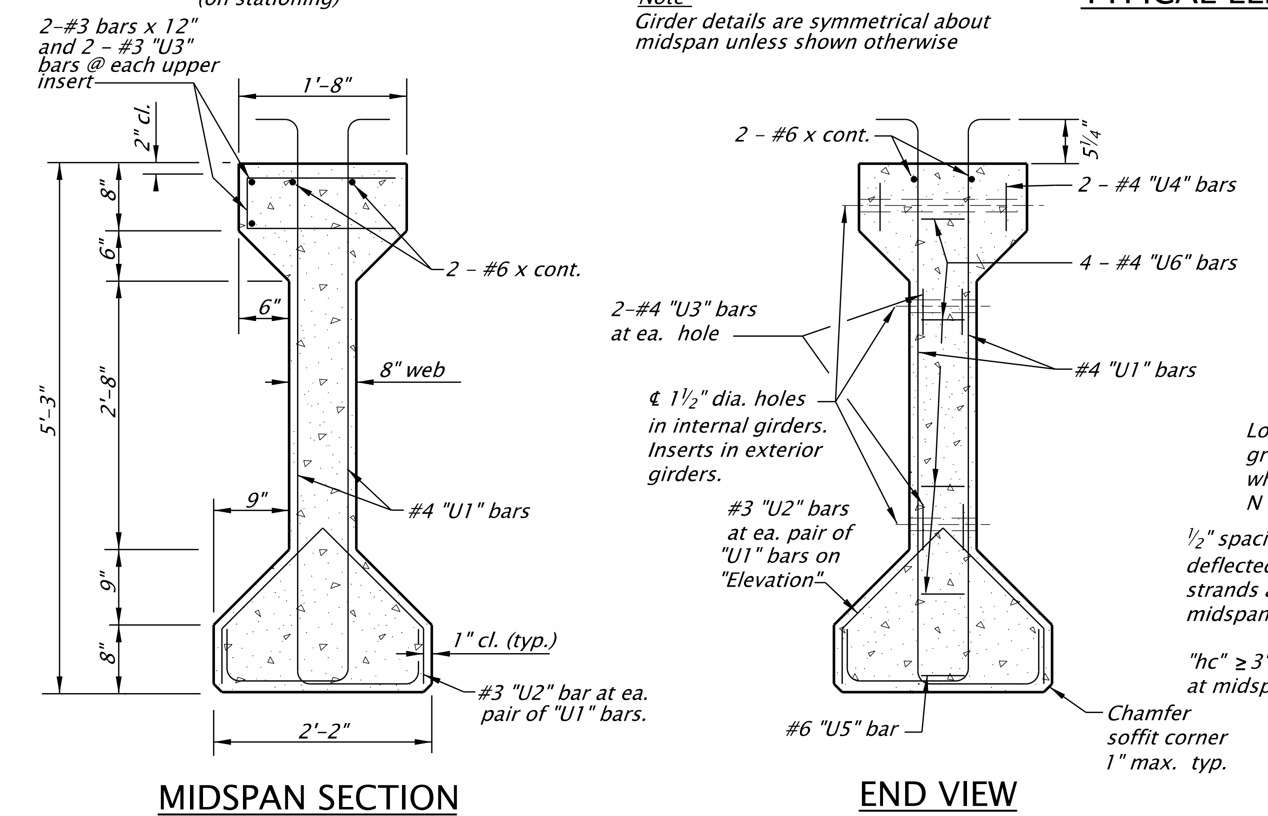
03-2017

br335.dgn

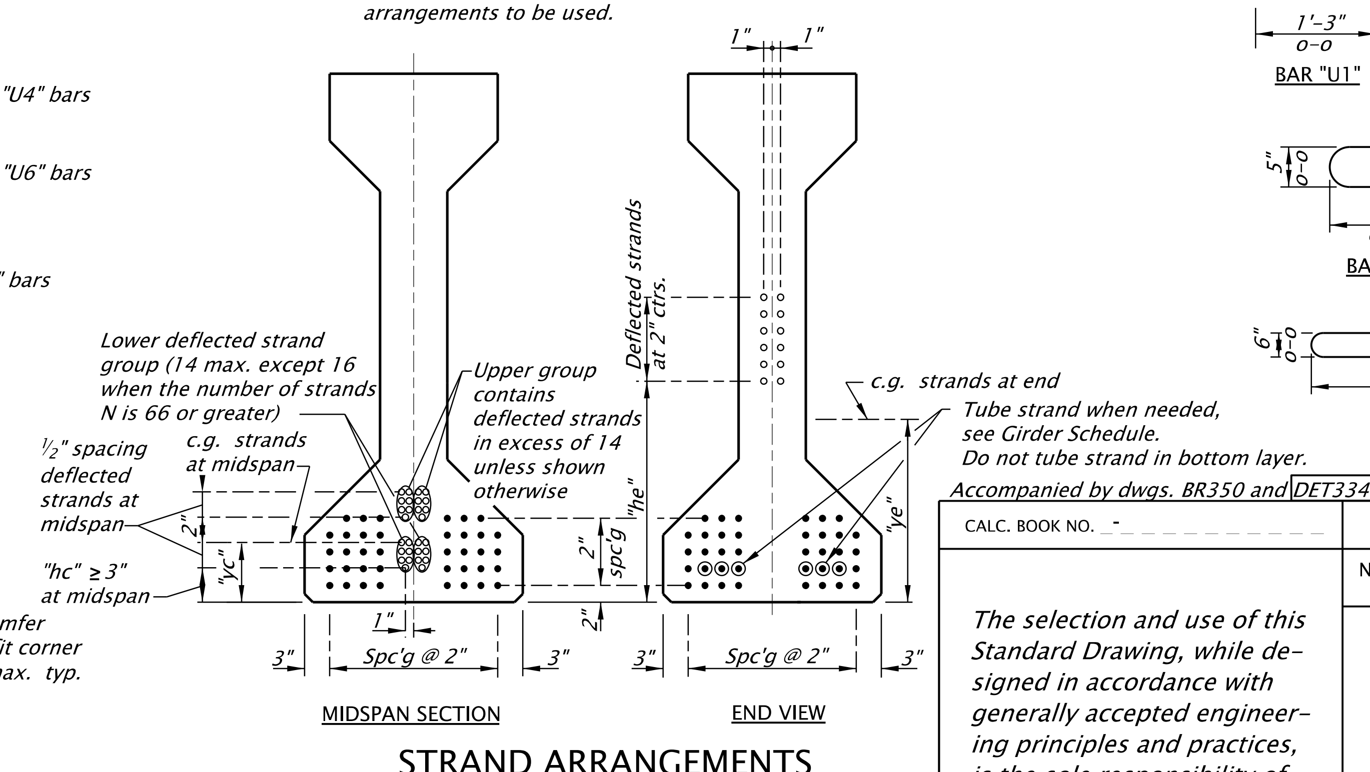
BR335



Note
Girder details are symmetrical about midspan unless shown otherwise



Note
These sections for dimensions only. See Figures on the Girder Schedule Sheet for the actual strand arrangements to be used.



Accompanied by dwgs. BR350 and DET3345

PRETENSIONING STEEL

Provide ½" dia. 7-wire low relaxation strands conforming to AASHTO M203 (ASTMA416), Grade 270, Supplement 1.

Provide an initial tension (after harping deflected strands) of 31.0 kips per strand unless shown otherwise. Deflected strands may be bundled between hold down points.

Provide the same tension as the straight strands after deflecting.

Detension strands alternately about the girder ¢ to limit prestress eccentricity.

MILD STEEL REINFORCEMENT

Provide reinforcing steel as specified in the General Notes for the project unless shown otherwise.

HANDLING PRESTRESSED CONCRETE GIRDERS

Keep girders in an upright position with the support points within 2'-0" of the girder ends.

Provide temporary diaphragm beams as required, see BR350.

DIAPHRAGM BEAMS

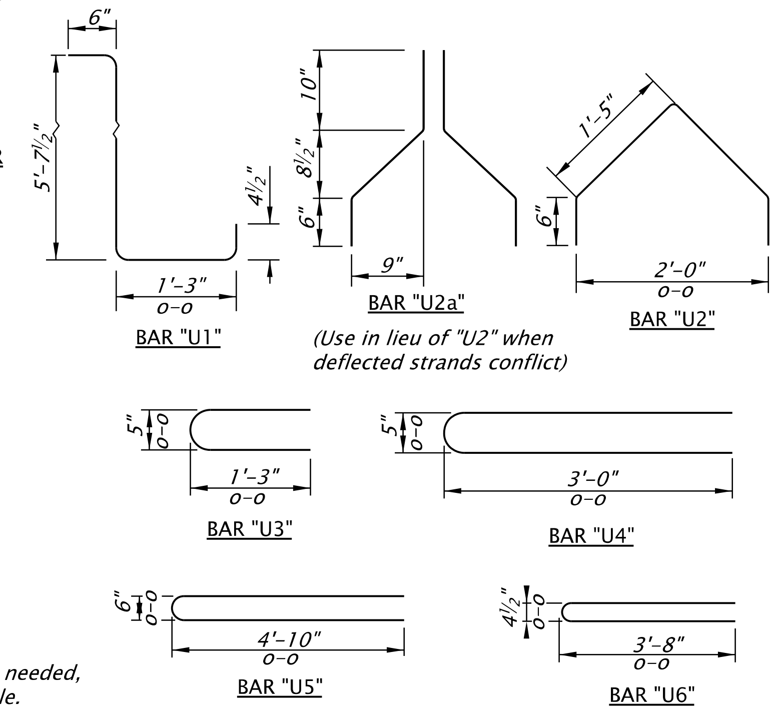
See structure plans for diaphragm beam locations.

Pour transverse beams a minimum of three days before deck pour.

Provide structural concrete inserts:

Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods AASHTO M314, Gr. 36 (ASTM A307) threaded full length.

Brace prestressed girders to prevent lateral deflection during diaphragm pour.



CALC. BOOK NO. -

SDR DATE: 20-April-2018

NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.

OREGON STANDARD DRAWINGS

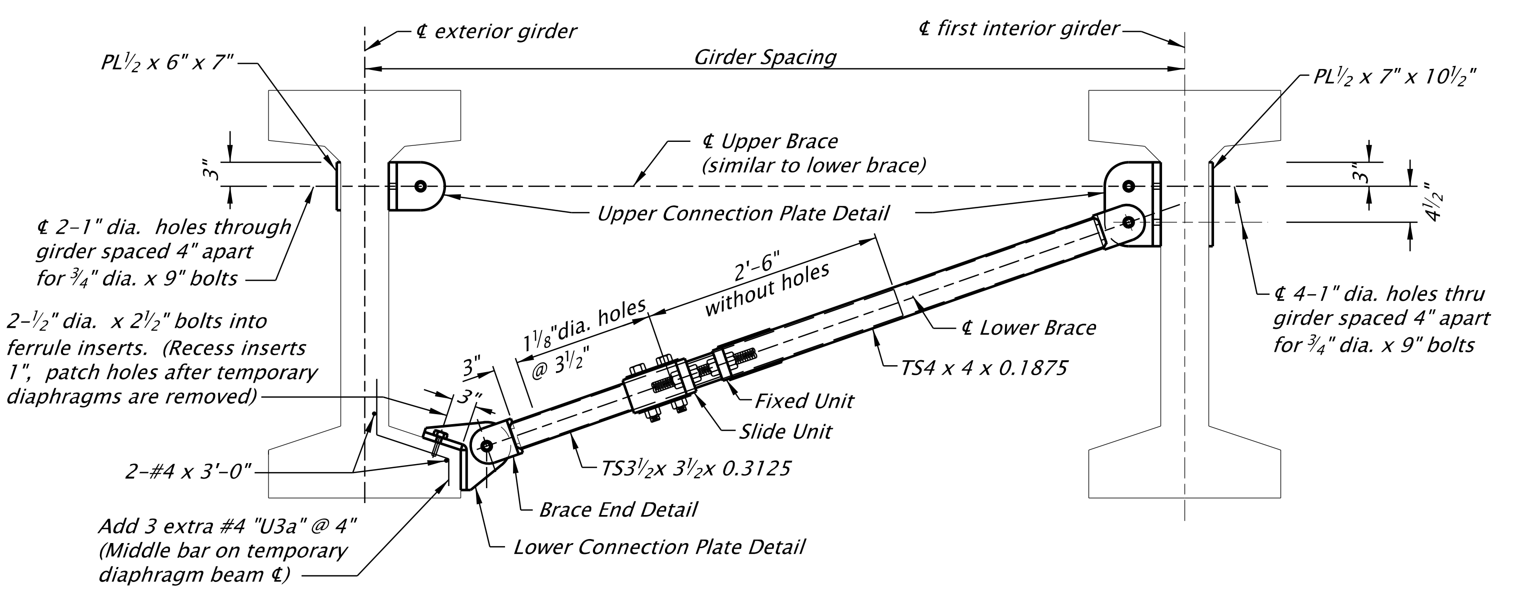
TYPE V - PRESTRESSED CONCRETE GIRDERS

2022

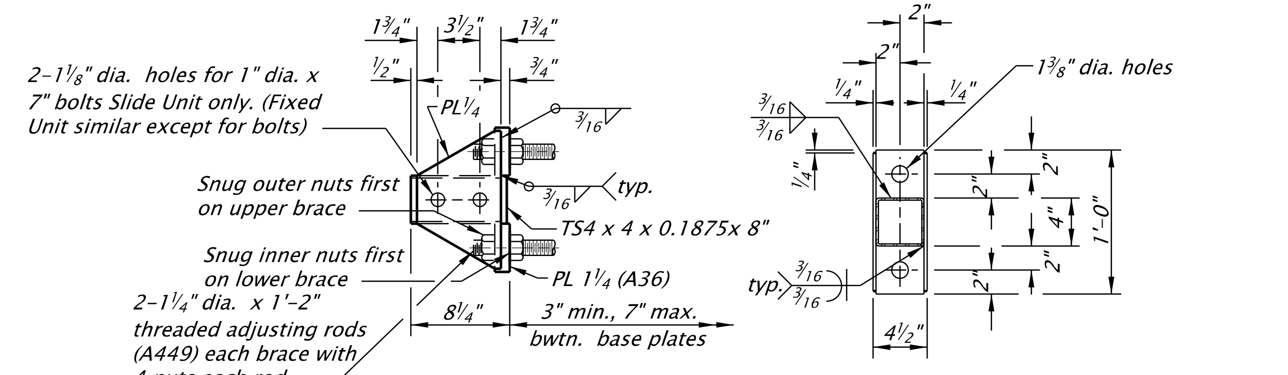
DATE	REVISION DESCRIPTION
-	-
-	-
-	-
-	-
-	-

br340.dgn 03-2017

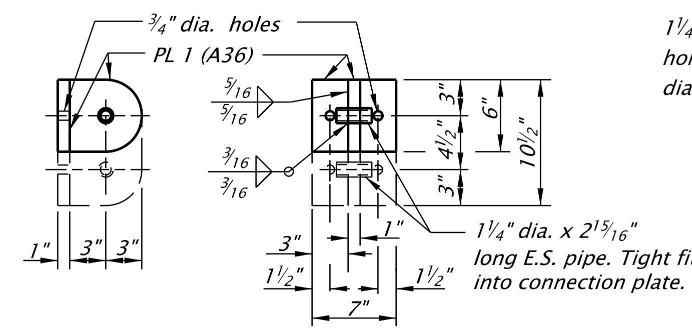
BR340



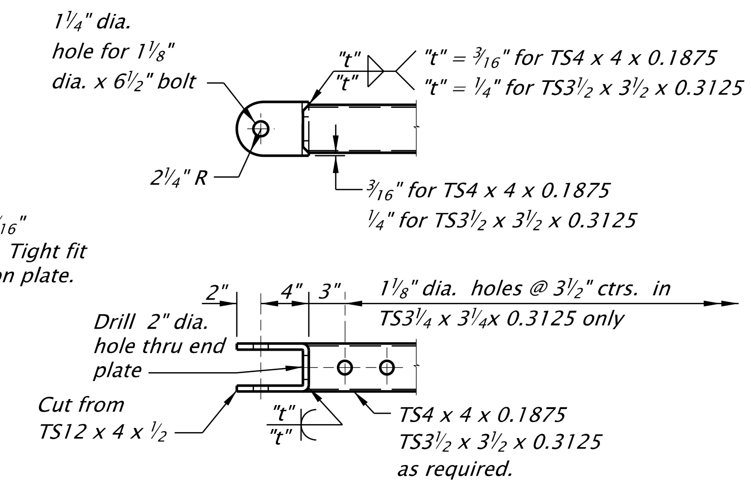
TEMPORARY DIAPHRAGM BEAM
(Bulb-I shown, other girders similar)



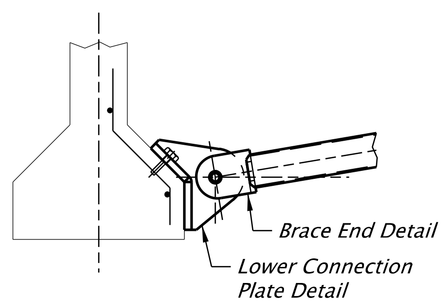
SLIDE UNIT DETAILS
FIXED UNIT SIMILAR



UPPER CONNECTION PLATE DETAIL



BRACE END DETAIL



LOWER CONNECTION DETAIL
TYPE II thru TYPE V GIRDERS

TEMPORARY DIAPHRAGM BEAM GENERAL NOTES

Install temporary diaphragm beams at exterior girders between end beams and permanent diaphragms prior to pouring permanent diaphragms and end beams. Space temporary diaphragms as needed to resist construction loads but not at more than 25 ft. apart. Assume the deck finishing machine carriage is at the most adverse position when computing construction loads. Remove temporary diaphragms after removing the deck overhang brackets.

Provide pipe conforming to ASTM Specification A53, Grade A or B. Provide structural tubing conforming to ASTM Specification A500, Grade A or B (or ASTM A513 with the strength and elongation characteristics of A500, Grade A or B).

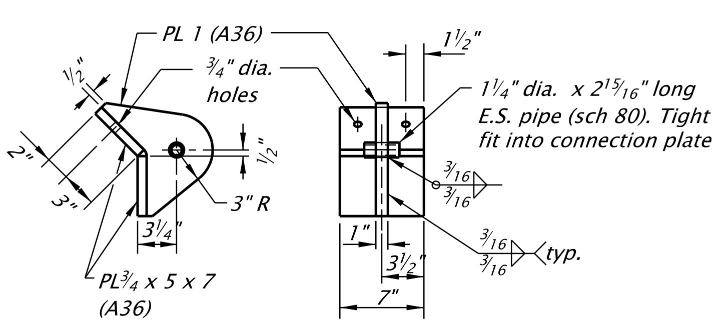
Provide other steel conforming to AASHTO Specification M183 (ASTM A36).

Provide mechanically galvanized high strength bolts conforming to ASTM Specification A325 unless shown otherwise. Provide mechanically galvanized threaded rods conforming to ASTM Specification A449.

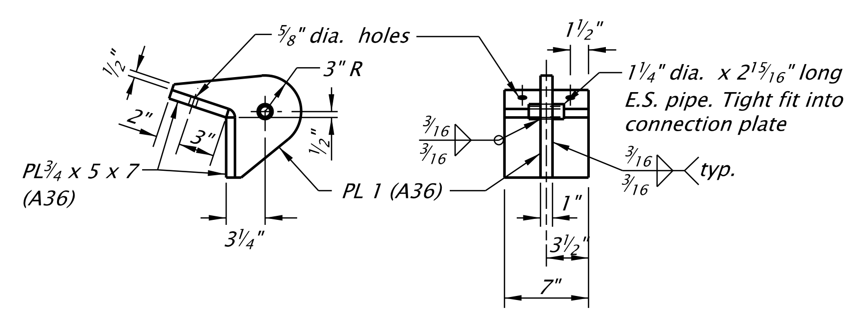
Tighten but do not torque high strength bolts and rods.

Hot-dip Galvanize all structural steel after fabrication.

Approved alternate designs prepared by a registered Professional Engineer may be used for temporary diaphragms.



LOWER CONNECTION PLATE DETAIL
TYPE II thru TYPE V GIRDERS



LOWER CONNECTION PLATE DETAIL
BULB-I and BULB-T GIRDERS

CALC. BOOK NO. -	SDR DATE: 20-April-2018
The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.	NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications **OREGON STANDARD DRAWINGS** **TEMPORARY DIAPHRAGM BEAM FOR PRESTRESSED CONCRETE GIRDERS** 2022

DATE	REVISION DESCRIPTION
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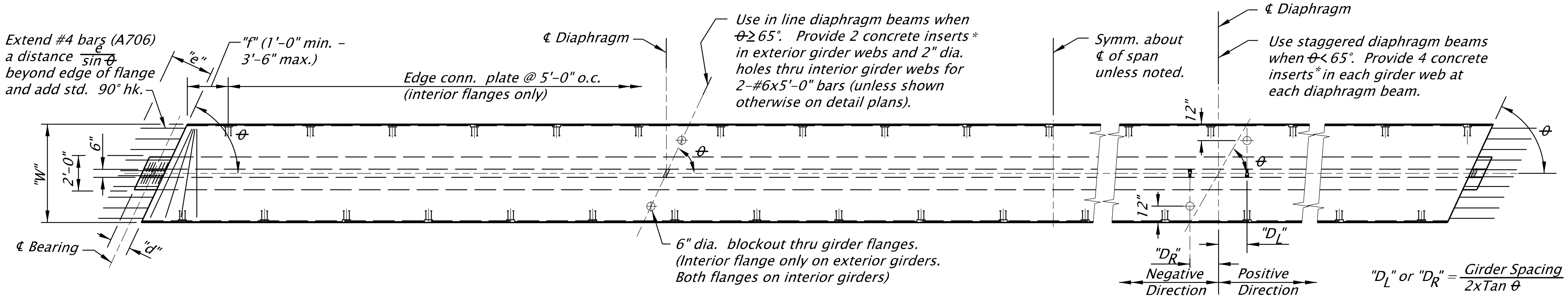
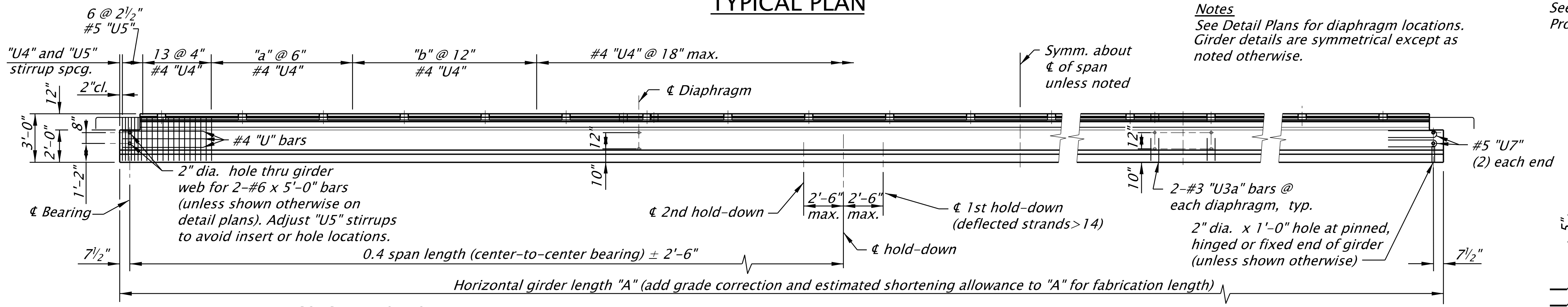
br350.dgn 03-2017

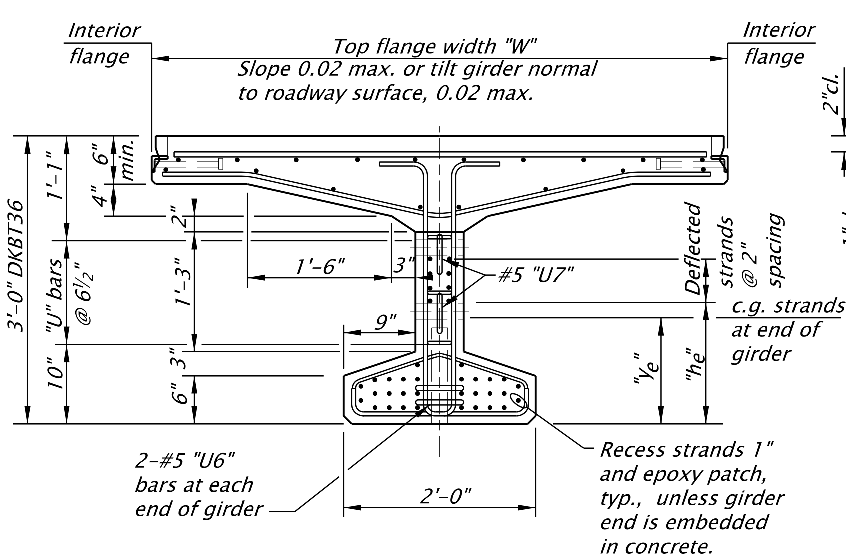


TYPICAL PLAN

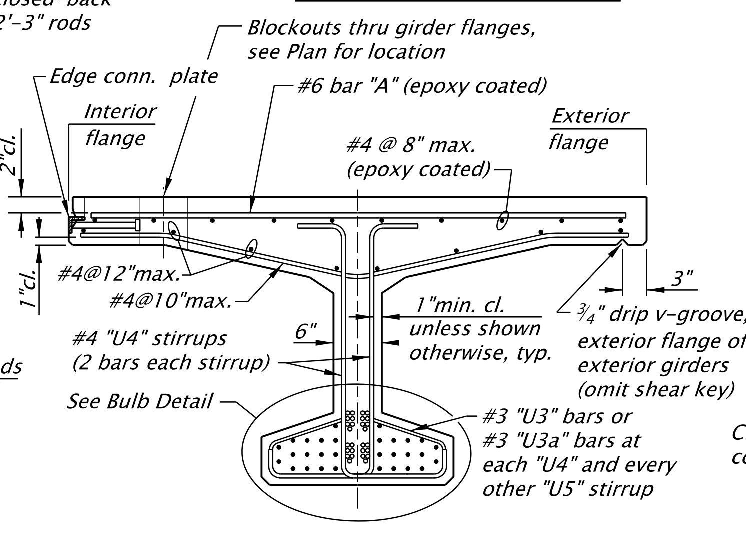


TYPICAL ELEVATION

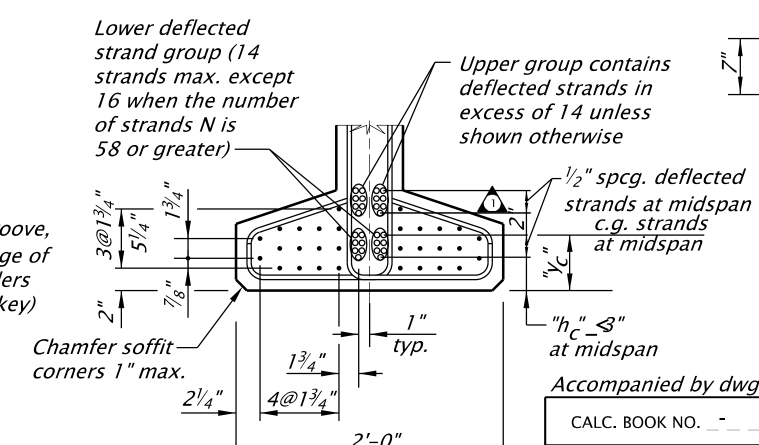
*CONCRETE INSERTS
Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods (ASTM A307) threaded full length.



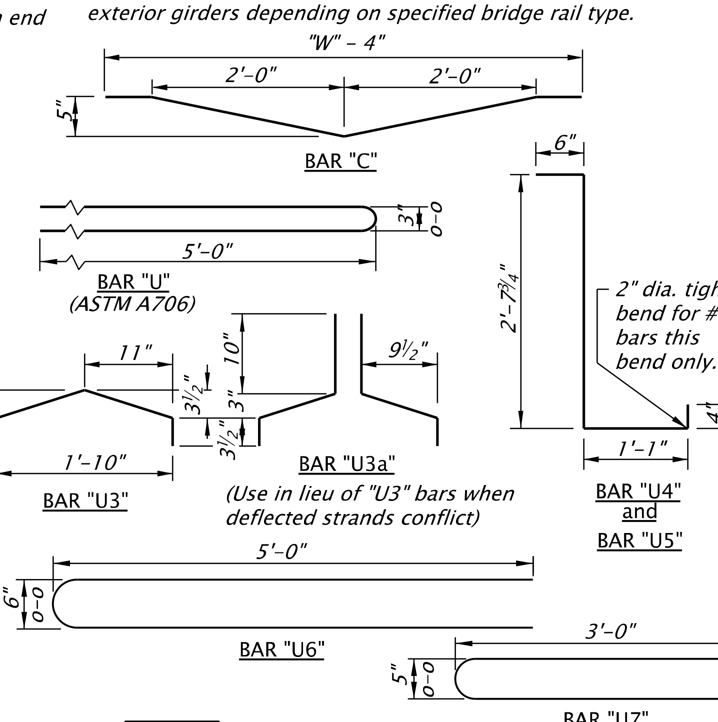
END VIEW



MIDSPAN SECTION



BULB DETAIL



Notes

Girders are designed for live loading and superimposed dead load as shown in the General Notes.
Provide the class of concrete shown on the Girder Schedule.
Provide concrete strength at transfer of prestress not less than that shown on the Girder Schedule.
Provide ½" dia. 7-wire low relaxation prestressing strands conforming to AASHTO M203 (ASTM A416) Grade 270, Supplement 1.
Tension straight strands initially to 31.0 kips, Ensure deflected strands have the same tension after deflecting.
Provide reinforcing steel as shown in the General Notes, except as noted.
Maintain girders in an upright position at all times with the support points within 2'-0" of the girder ends.
Provide a transverse broom finish on tops of girders.
Provide general finish for all other surfaces, except as shown in the General Notes.
Deflected strands may be bundled between hold-down points.
Provide temporary diaphragm beams as required, see BR350.
See dwg. $\boxed{\text{DET3380}}$ for "a", "b", "d", "e", "f" and "θ" values.
Provide welded wire fabric conforming to ASTM A497.

MINIMUM DECK REINFORCEMENT FOR INTERIOR GIRDERS			
BAR	TOP FLANGE WIDTH "W" (in)		
	60 to 72	72 to 84	84 to 102
"A" Spacing (in)	7	6	5

Additional top deck reinforcement may be required for exterior girders depending on specified bridge rail type.

Accompanied by dwgs. BR350 and $\boxed{\text{DET3380}}$

CALC. BOOK NO. -

SDR DATE: 20-April-2018

NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without consulting a Registered Professional Engineer.

OREGON STANDARD DRAWINGS

36" DECK BULB-T (PRECAST PRESTRESSED CONCRETE) GIRDER

2022

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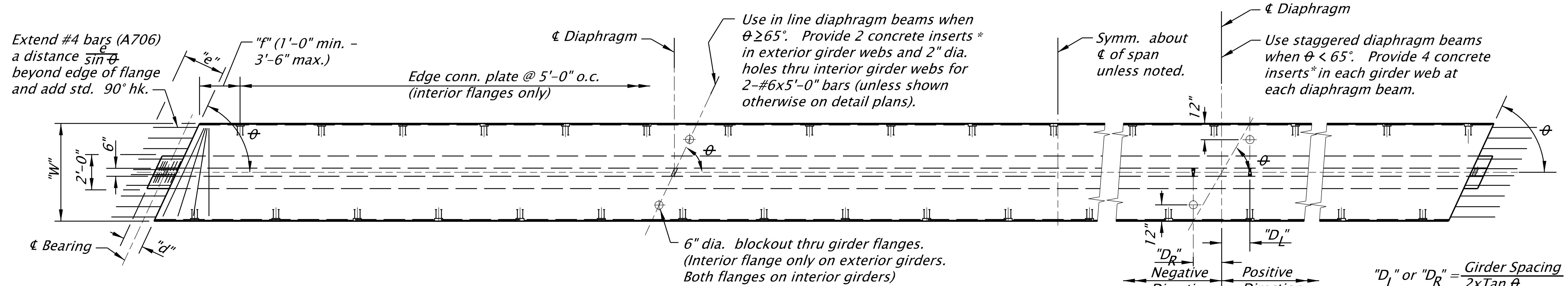
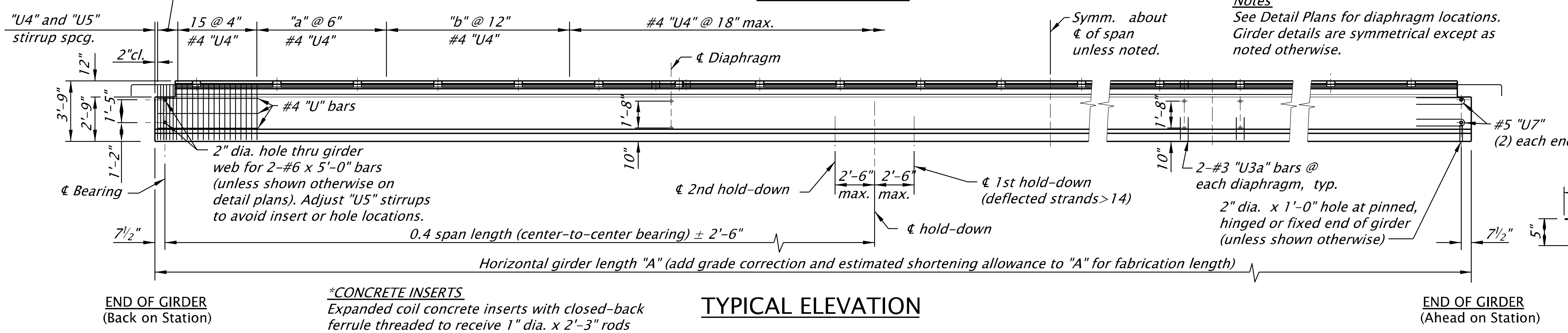
br360.dgn 03-2017

BR360



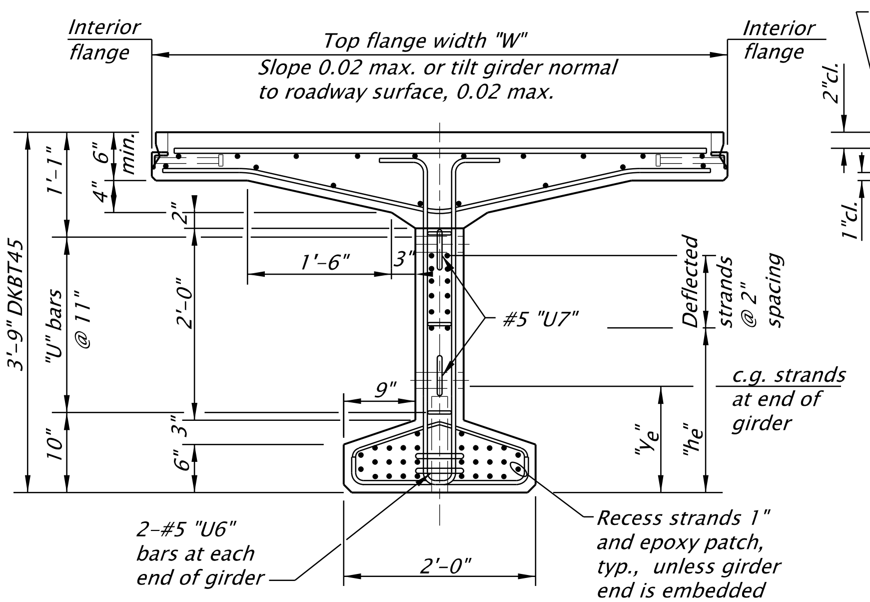
TYPICAL PLAN



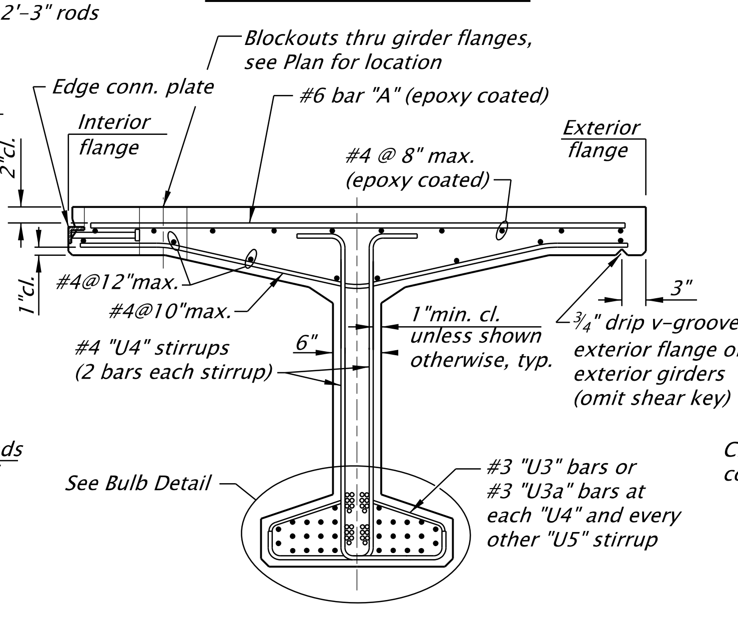
Notes
See Detail Plans for diaphragm locations.
Girder details are symmetrical except as noted otherwise.

TYPICAL ELEVATION

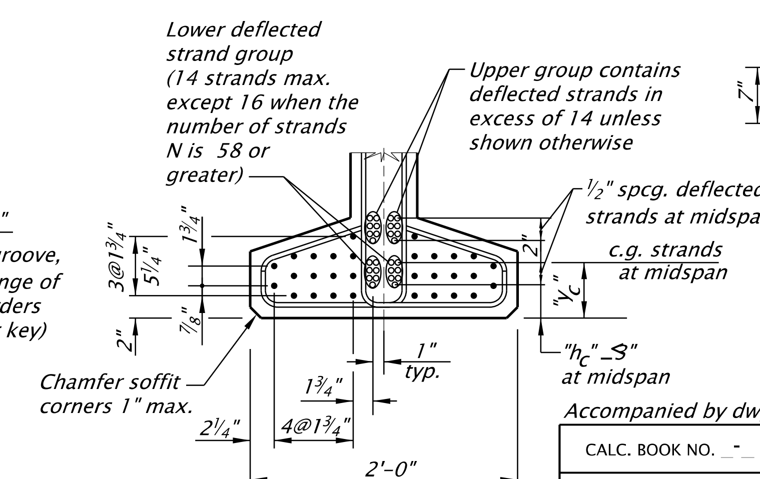
*CONCRETE INSERTS
Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods (ASTM A307) threaded full length.



END VIEW



MIDSPAN SECTION

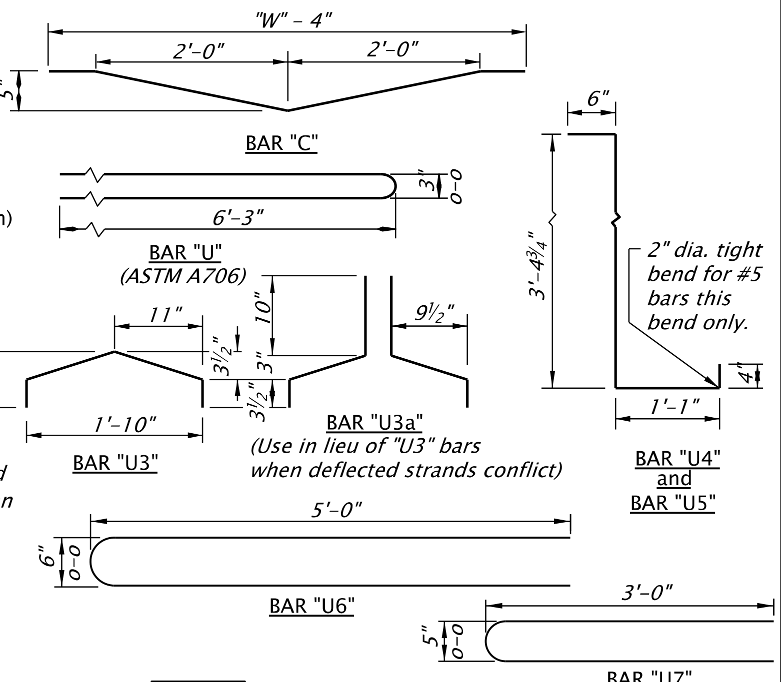


BULB DETAIL

Notes
Girders are designed for live loading and superimposed dead load as shown in the General Notes.
Provide the class of concrete shown on the Girder Schedule.
Provide concrete strength at transfer of prestress not less than that shown on the Girder Schedule.
Provide ½" dia. 7-wire low relaxation prestressing strands conforming to AASHTO M203 (ASTM A416) Grade 270, Supplement 1.
Tension straight strands initially to 31.0 kips, Ensure deflected strands have the same tension after deflecting.
Provide reinforcing steel as shown in the General Notes, except as noted.
Maintain girders in an upright position at all times with the support points within 2'-0" of the girder ends.
Provide a transverse broom finish on tops of girders.
Provide general finish for all other surfaces, except as shown in the General Notes.
Deflected strands may be bundled between hold-down points.
Provide temporary diaphragm beams as required, see BR350.
See dwg. DET3380 for "a", "b", "d", "e", "f" and "θ" values.
Provide welded wire fabric conforming to ASTM A497.

MINIMUM DECK REINFORCEMENT FOR INTERIOR GIRDERS			
BAR	TOP FLANGE WIDTH "W" (in)		
	60 to 72	72 to 84	84 to 102
"A" Spacing (in)	7	6	5

Additional top deck reinforcement may be required for exterior girders depending on specified bridge rail type.



Accompanied by dwgs. BR350 and DET3380

CALC. BOOK NO. -

SDR DATE: 20-April-2018

NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

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OREGON STANDARD DRAWINGS

45" DECK BULB-T (PRECAST PRESTRESSED CONCRETE) GIRDER

2022

DATE	REVISION DESCRIPTION
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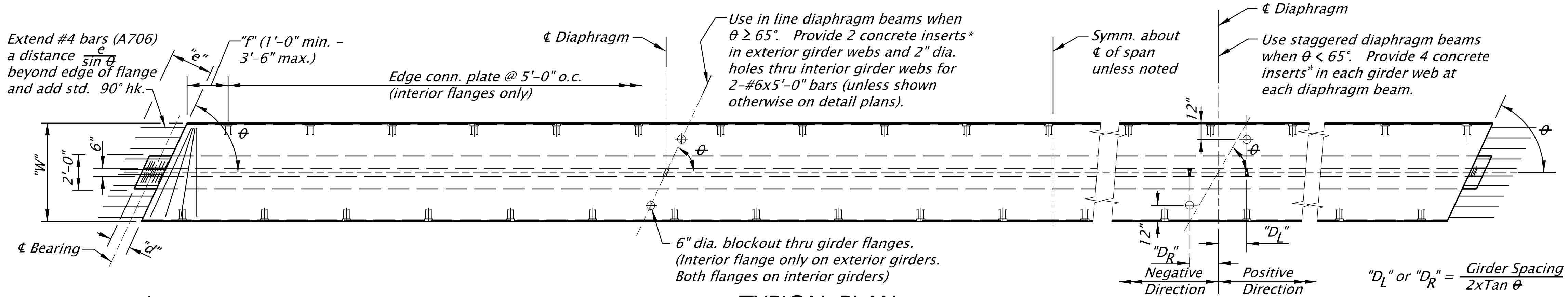
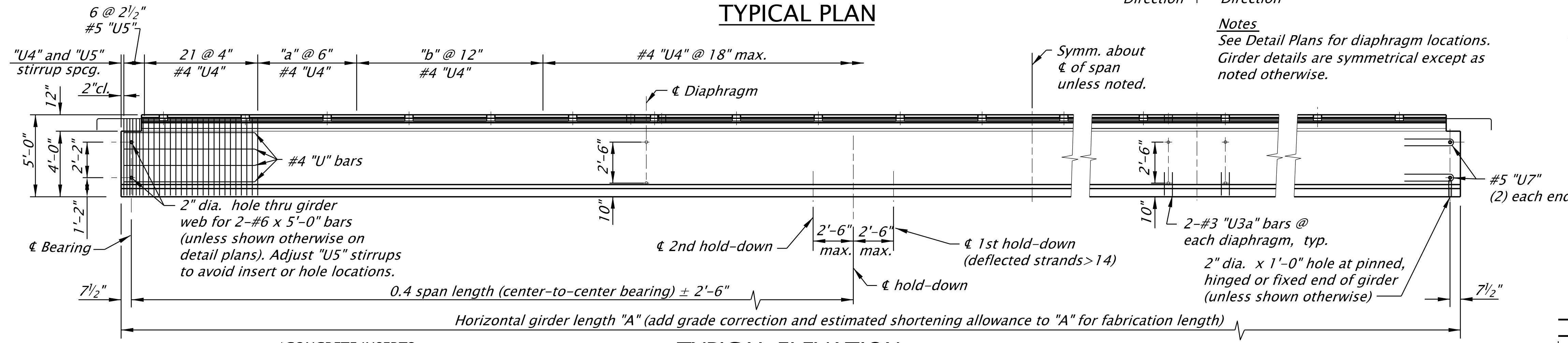
br365.dgn 03-2017

BR365

Effective Date: December 1, 2022 - May 31, 2023



TYPICAL PLAN



TYPICAL ELEVATION

END OF GIRDER (Back on Station)

END OF GIRDER (Ahead on Station)

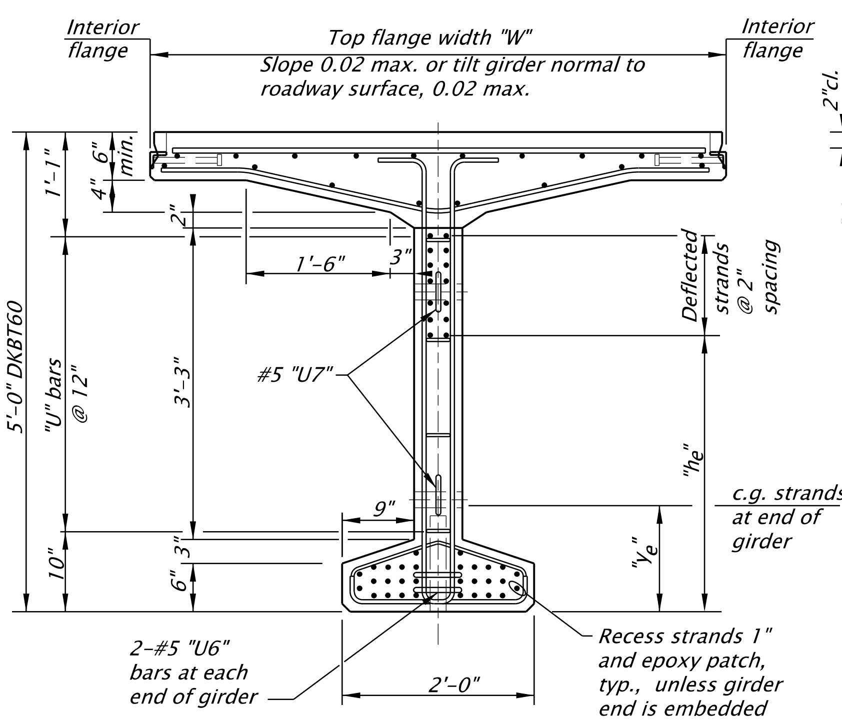
*CONCRETE INSERTS
Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods (ASTM A307) threaded full length.

Notes
See Detail Plans for diaphragm locations.
Girder details are symmetrical except as noted otherwise.

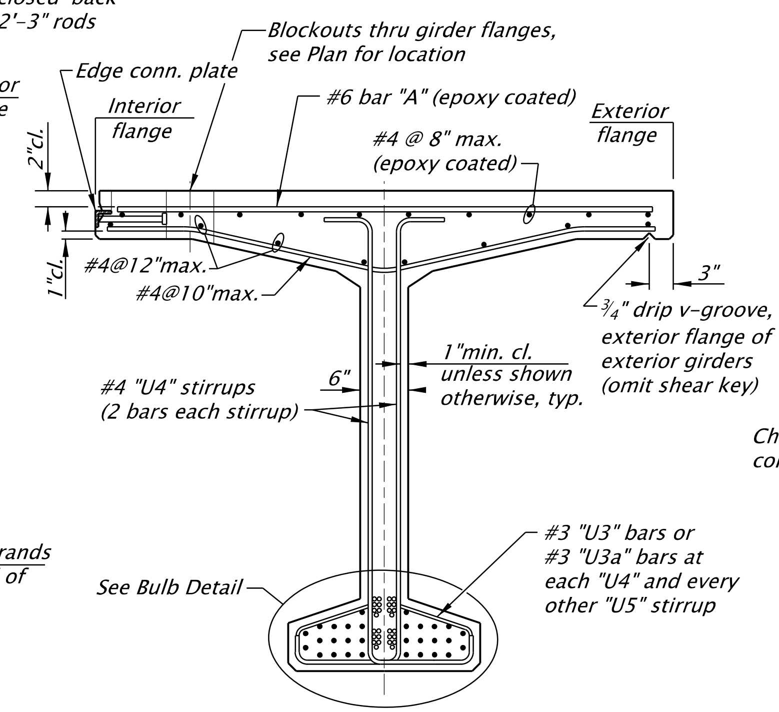
Notes
Girders are designed for live loading and superimposed dead load as shown in the General Notes.
Provide the class of concrete shown on the Girder Schedule.
Provide concrete strength at transfer of prestress not less than that shown on the Girder Schedule.
Provide ½" dia. 7-wire low relaxation prestressing strands conforming to AASHTO M203 (ASTM A416) Grade 270, Supplement 1.
Tension straight strands initially to 31.0 kips, Ensure deflected strands have the same tension after deflecting.
Provide reinforcing steel as shown in the General Notes, except as noted.
Maintain girders in an upright position at all times with the support points within 2'-0" of the girder ends.
Provide a transverse broom finish on tops of girders.
Provide general finish for all other surfaces, except as shown in the General Notes.
Deflected strands may be bundled between hold-down points.
Provide temporary diaphragm beams as required, see BR350.
See dwg. DET3380 for "a", "b", "d", "e", "f" and "θ" values.
Provide welded wire fabric conforming to ASTM A497.

MINIMUM DECK REINFORCEMENT FOR INTERIOR GIRDERS			
BAR	TOP FLANGE WIDTH "W" (in)		
	60 to 72	72 to 84	84 to 102
"A" Spacing (in)	7	6	5

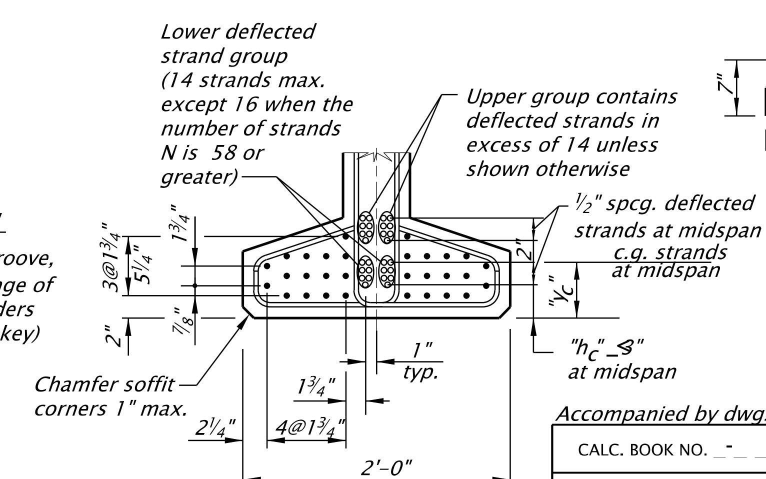
Additional top deck reinforcement may be required for exterior girders depending on specified bridge rail type.



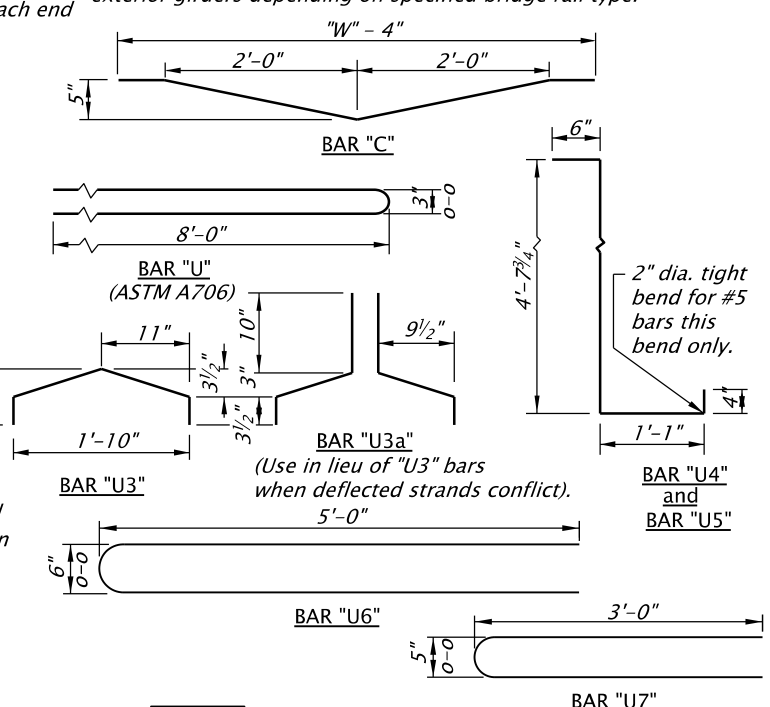
END VIEW



MIDSPAN SECTION



BULB DETAIL



Accompanied by dwgs. BR350 and DET3380

CALC. BOOK NO. -	SDR DATE: 20-April-2018

NOTE: All material and workmanship shall be in accordance with the current Oregon Standard Specifications

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OREGON STANDARD DRAWINGS

60" DECK BULB-T (PRECAST PRESTRESSED CONCRETE) GIRDER

2022

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